# Getting to Know Libero

## Course Workbook

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# About this Workbook

This workbook is designed to be used in conjunction with the Getting to Know Libero workbook.

The contents of this workbook are created by Adiuvo Engineering & Training, Ltd.

If you have any questions about the contents, or need assistance, please contact Adam Taylor at adam@adiuvoengineering.com.

# Pre-Lab

## Workshop Pre-requisites

# Required Hardware

There is no required hardware for this course.

# Downloads and Installations

**Step 1 –** Download and install the following at least one day prior to the workshop. This may take a significant amount of time and drive space.

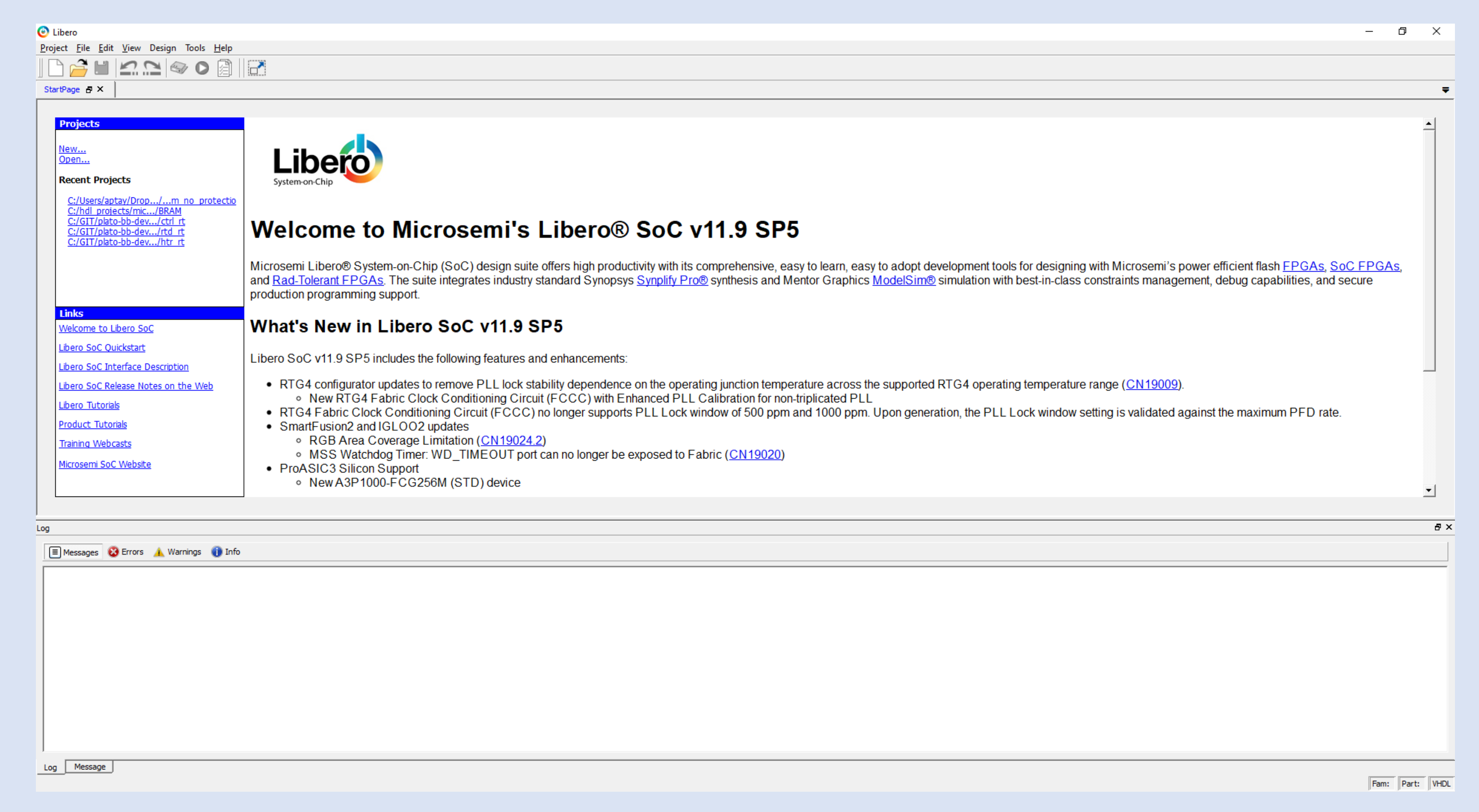
| | |
|---|---|
| Libero SoC 11.9 | Download |
| Source Project Files | Download |

# Lab 1

## Overview and Introduction to Libero

# Lab 1: Overview and Introduction to Libero

**Step 1 –** Open Libero 11.9



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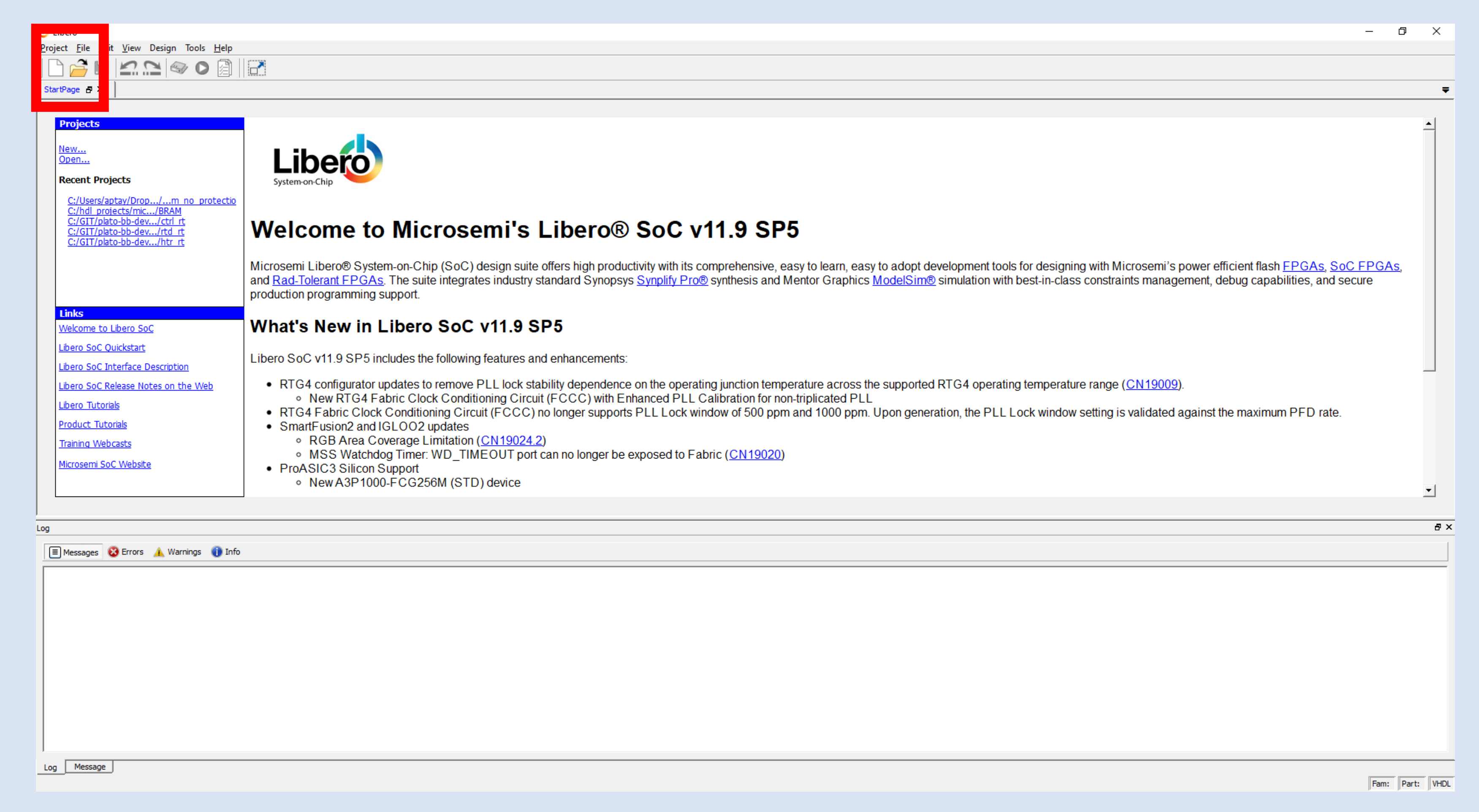
**Step 2 –** Create a new project from the project menu



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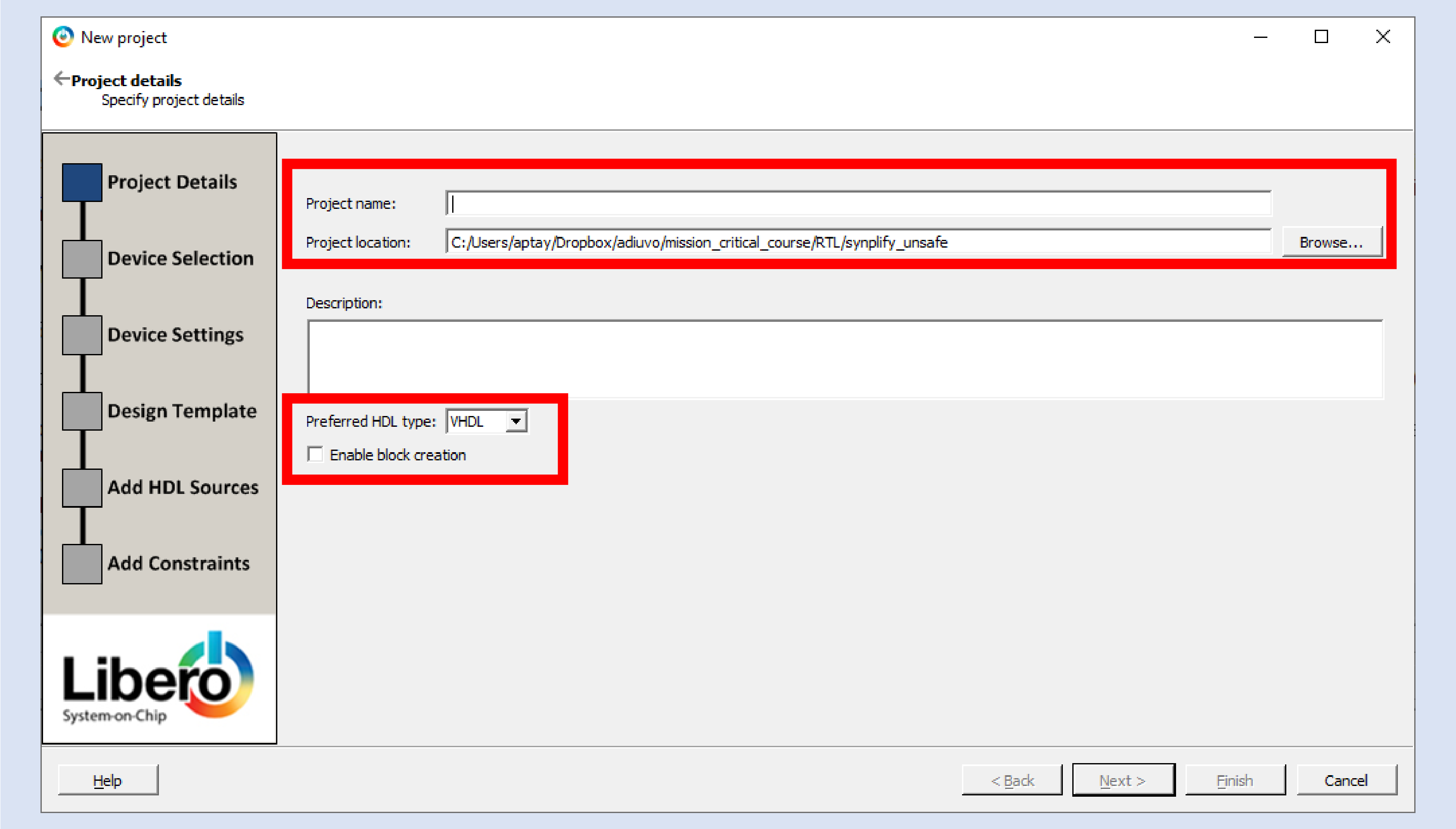
**Step 3 –** Enter a project name and location – select the HDL type as VHDL – leave block creation unchecked



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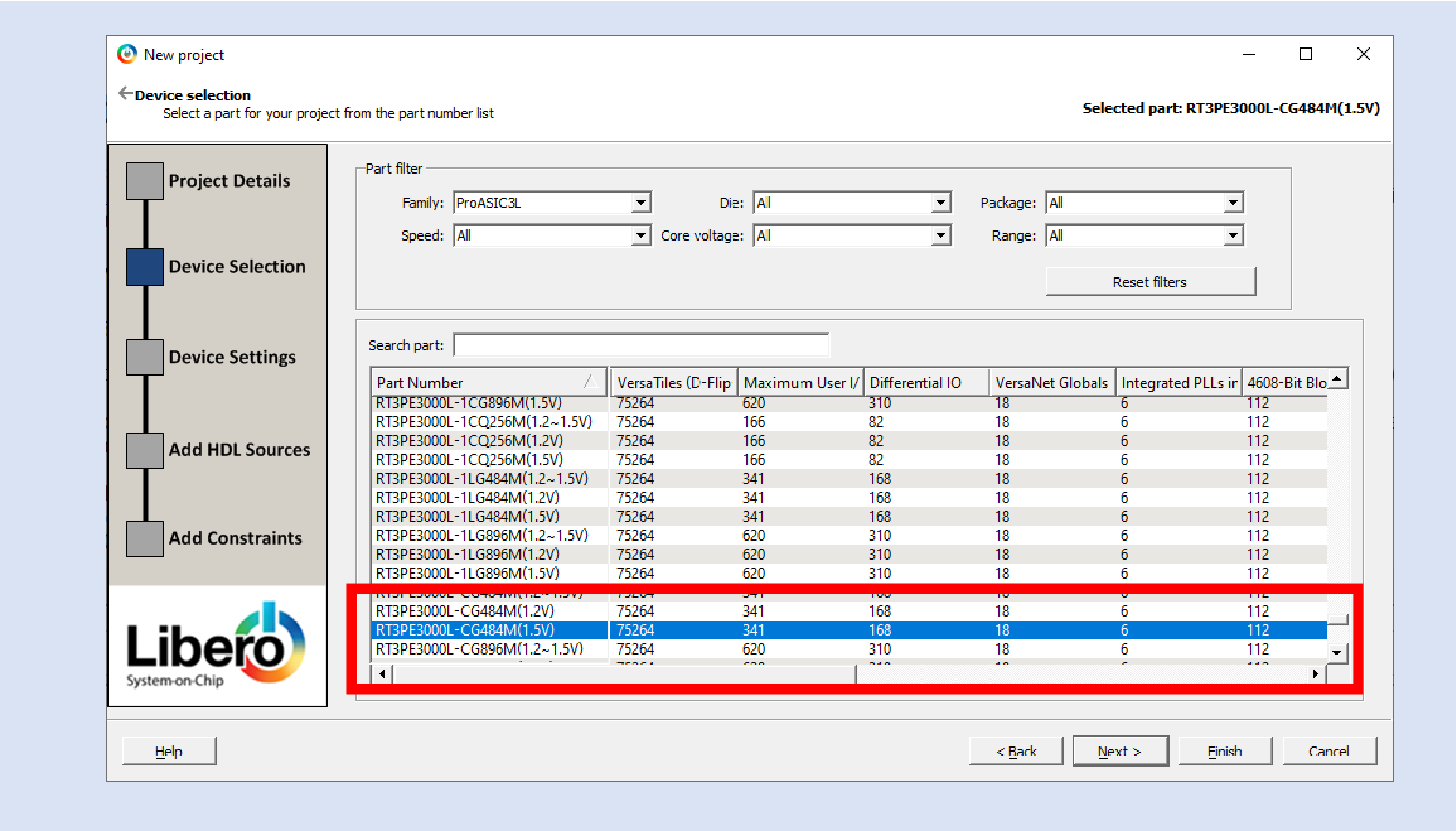
**Step 4 –** Select a RT3PE3000L device



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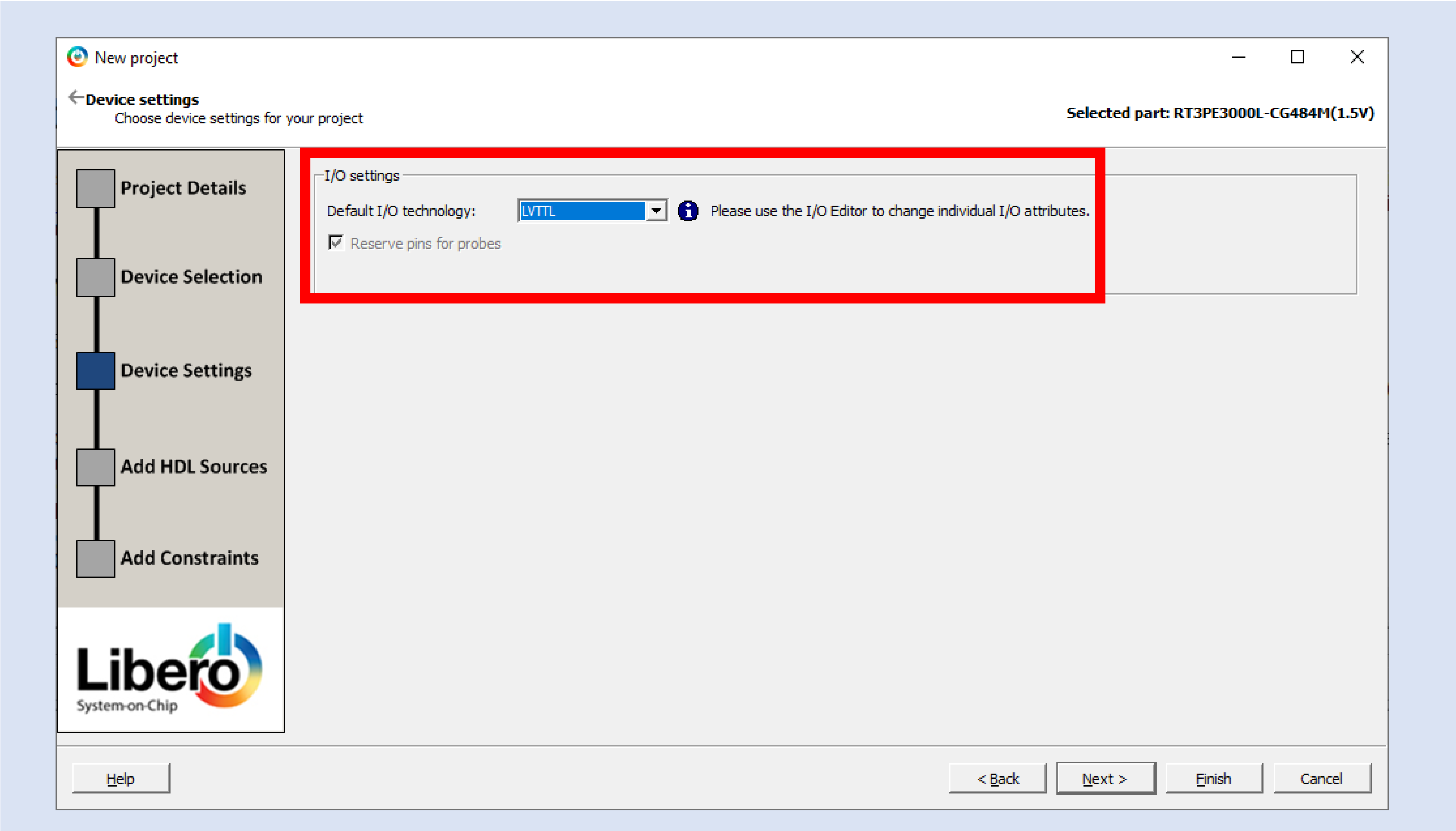
**Step 5 –** Select LVTTL as the Default I/O Technology



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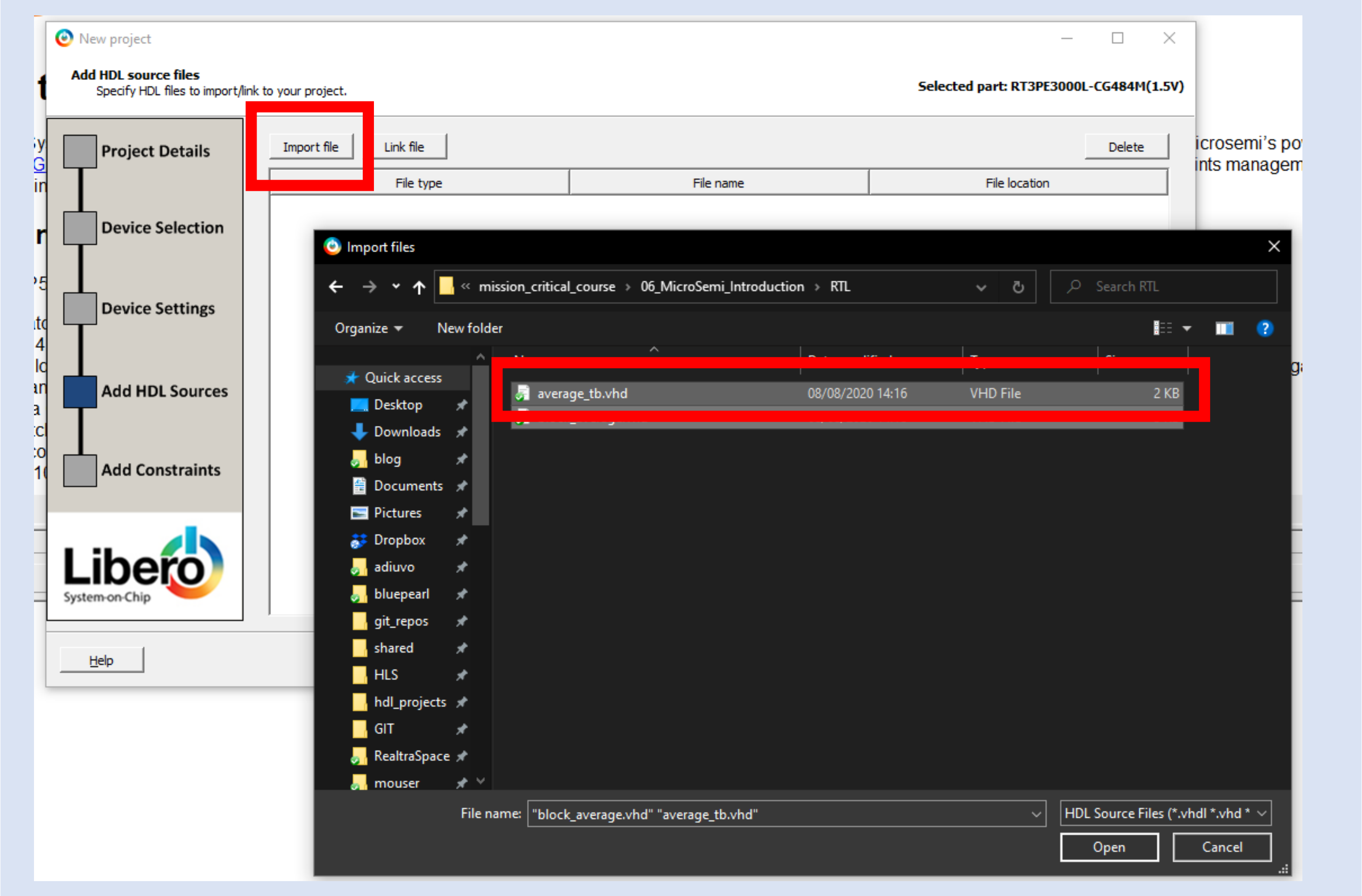
**Step 6 –** Select import file and import the block_average.vhd example file provided



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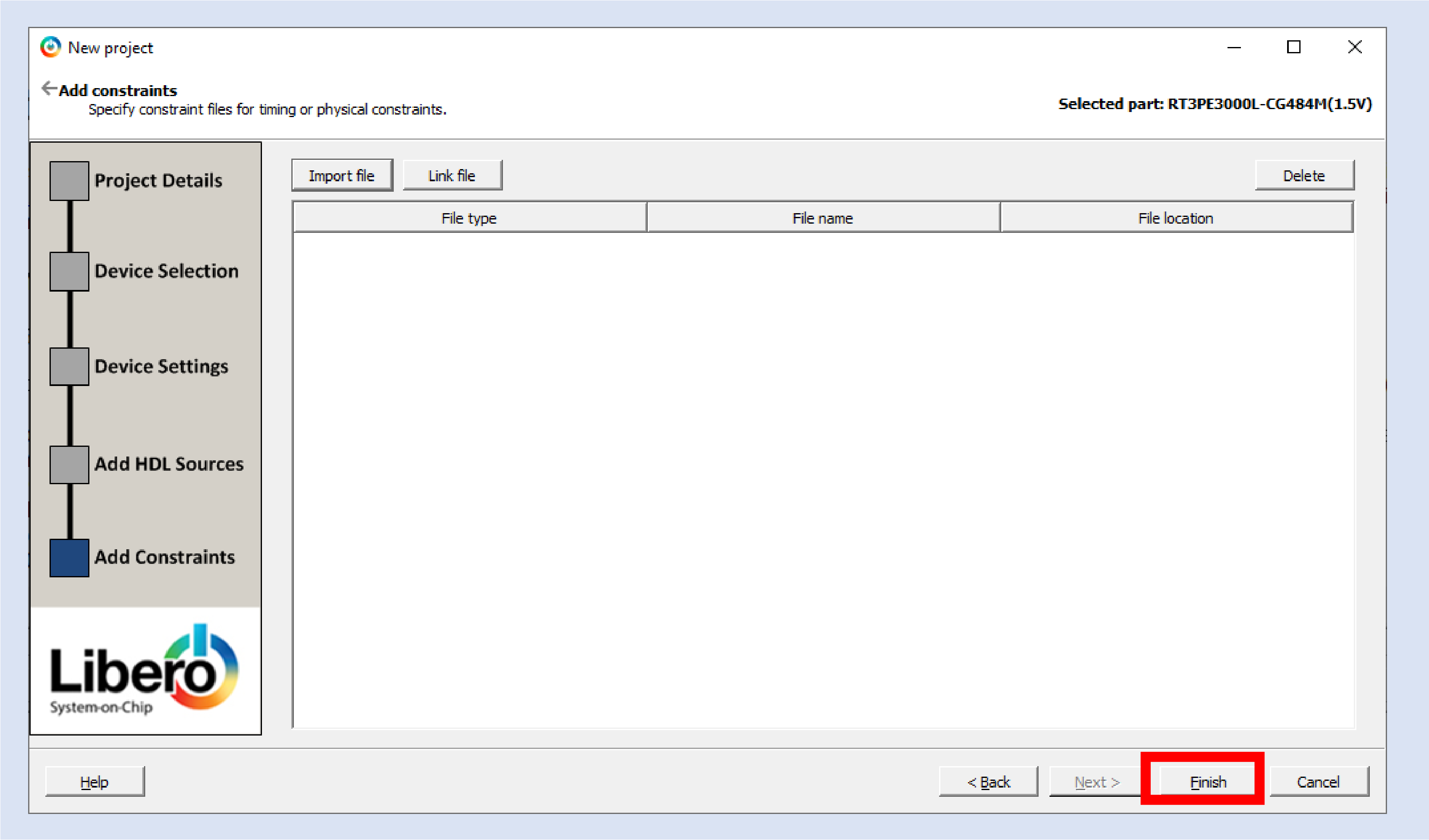
**Step 7 –** Leave the constraint empty and select finish



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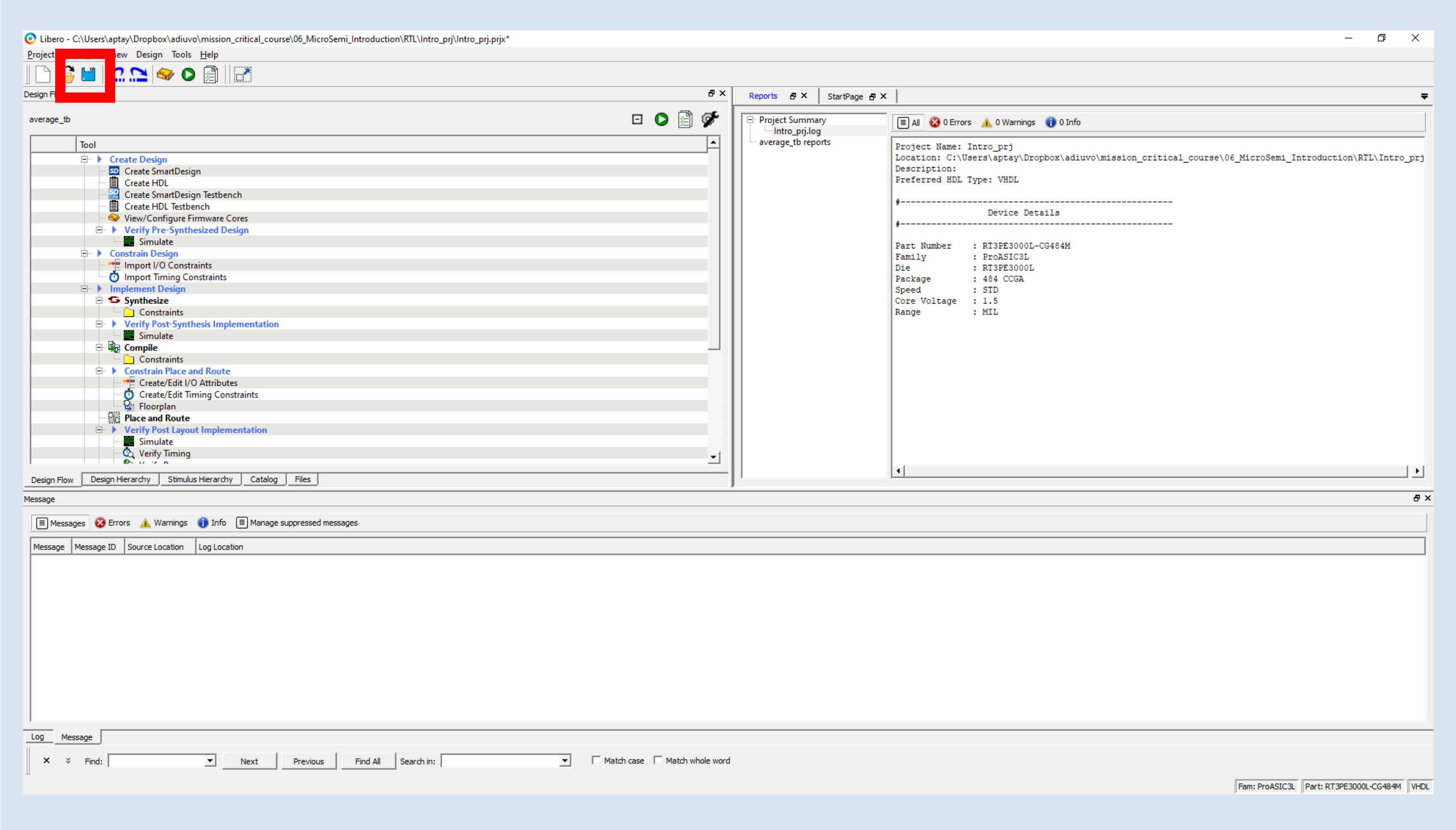
**Step 8 –** This will open the project in Libero – Click on Save



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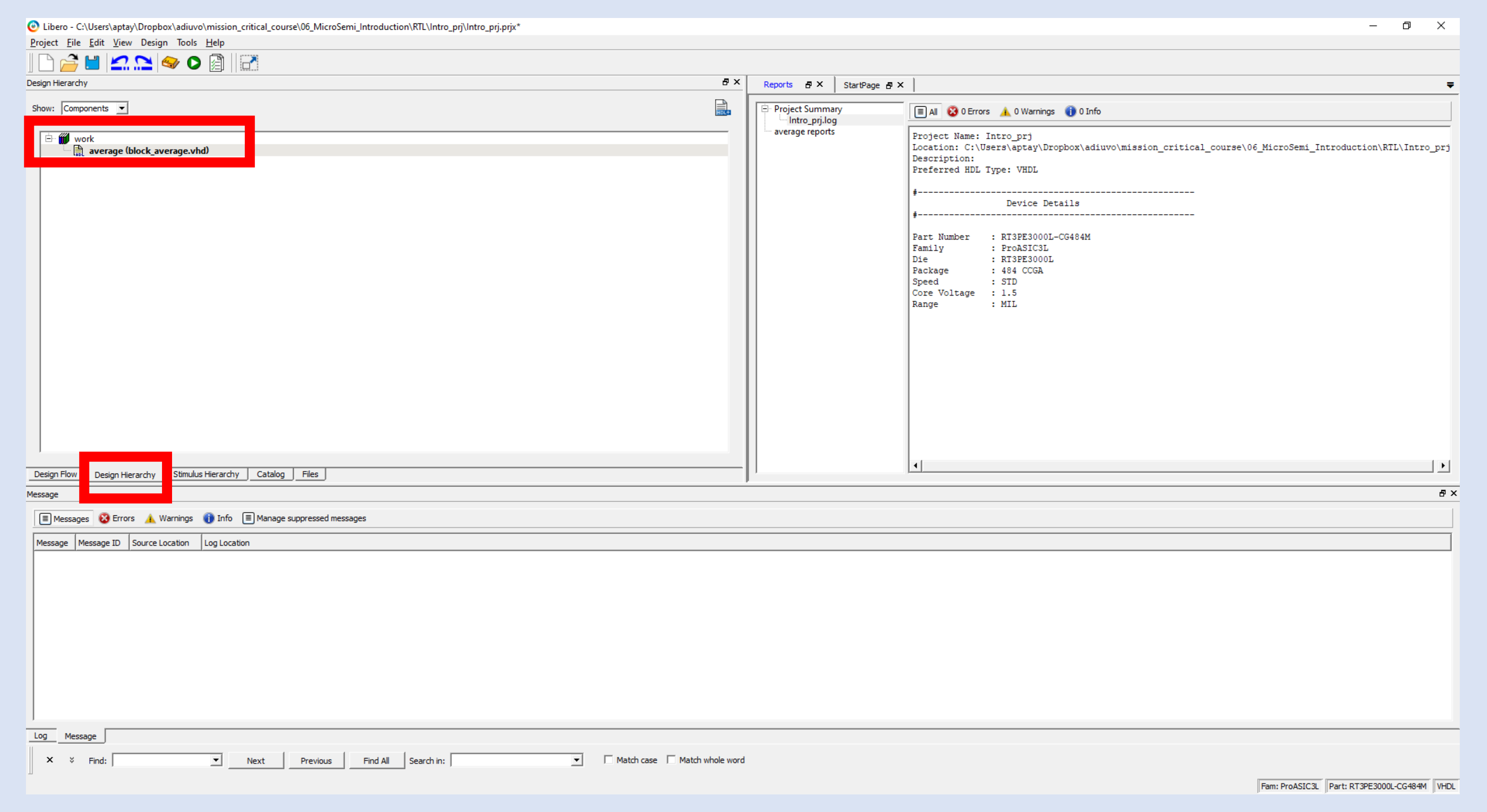
**Step 9 –** Select Design Hierarchy – You will should see the RTL file



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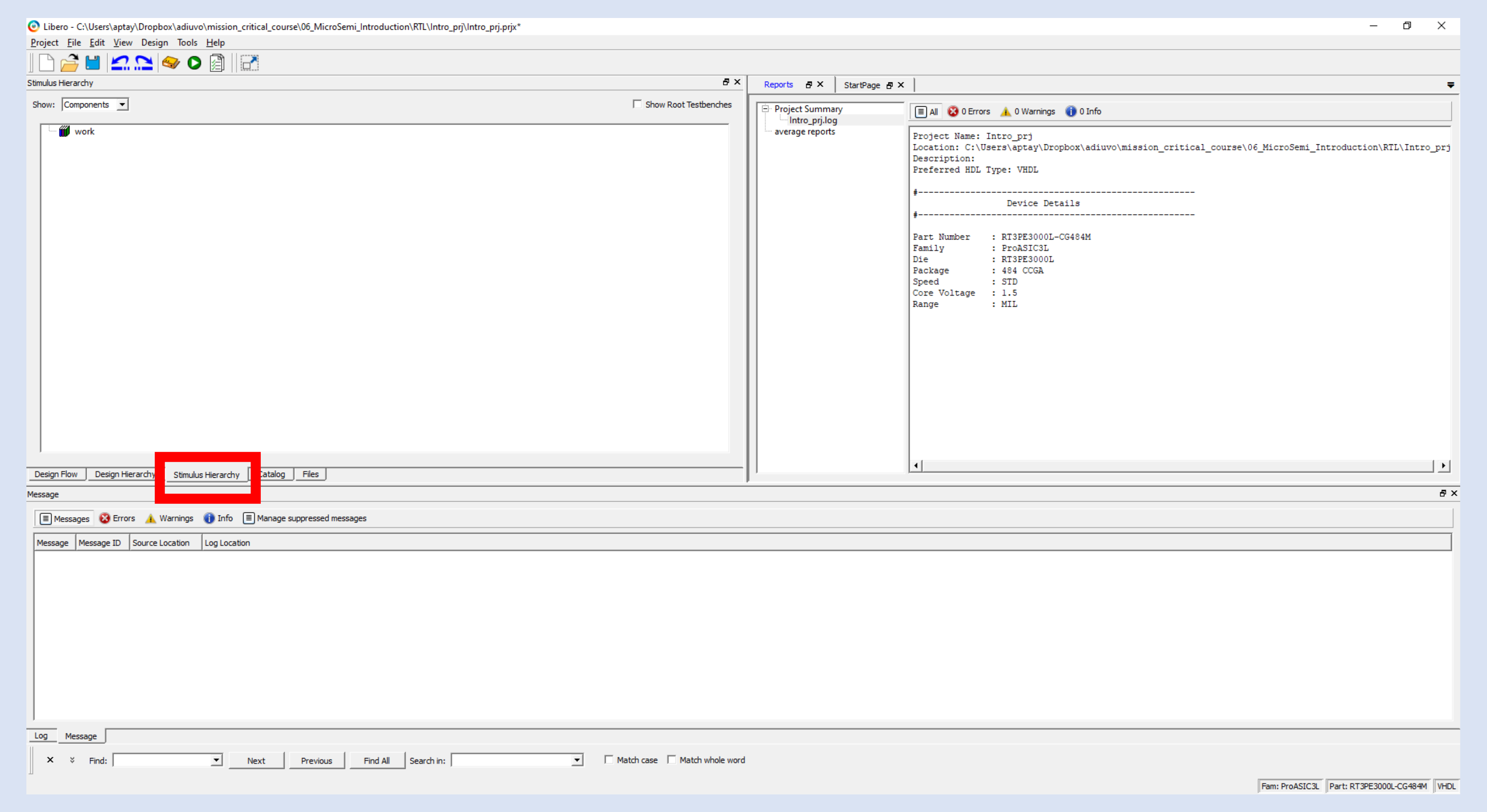
**Step 10 –** Select the stimulus hierarchy tab – notice it is empty



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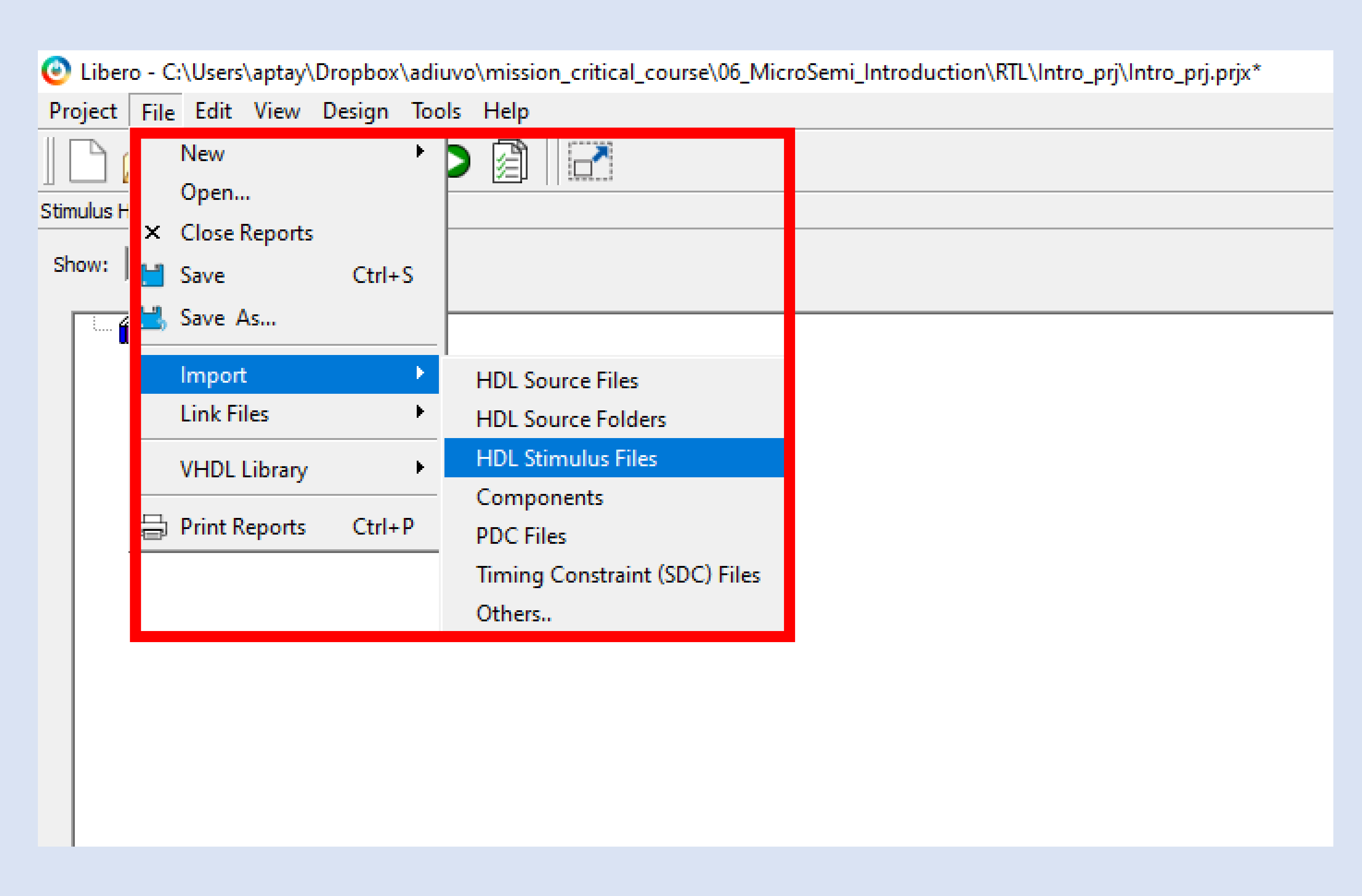
**Step 11 –** Select File -> Import -> HDL Stimulus Files



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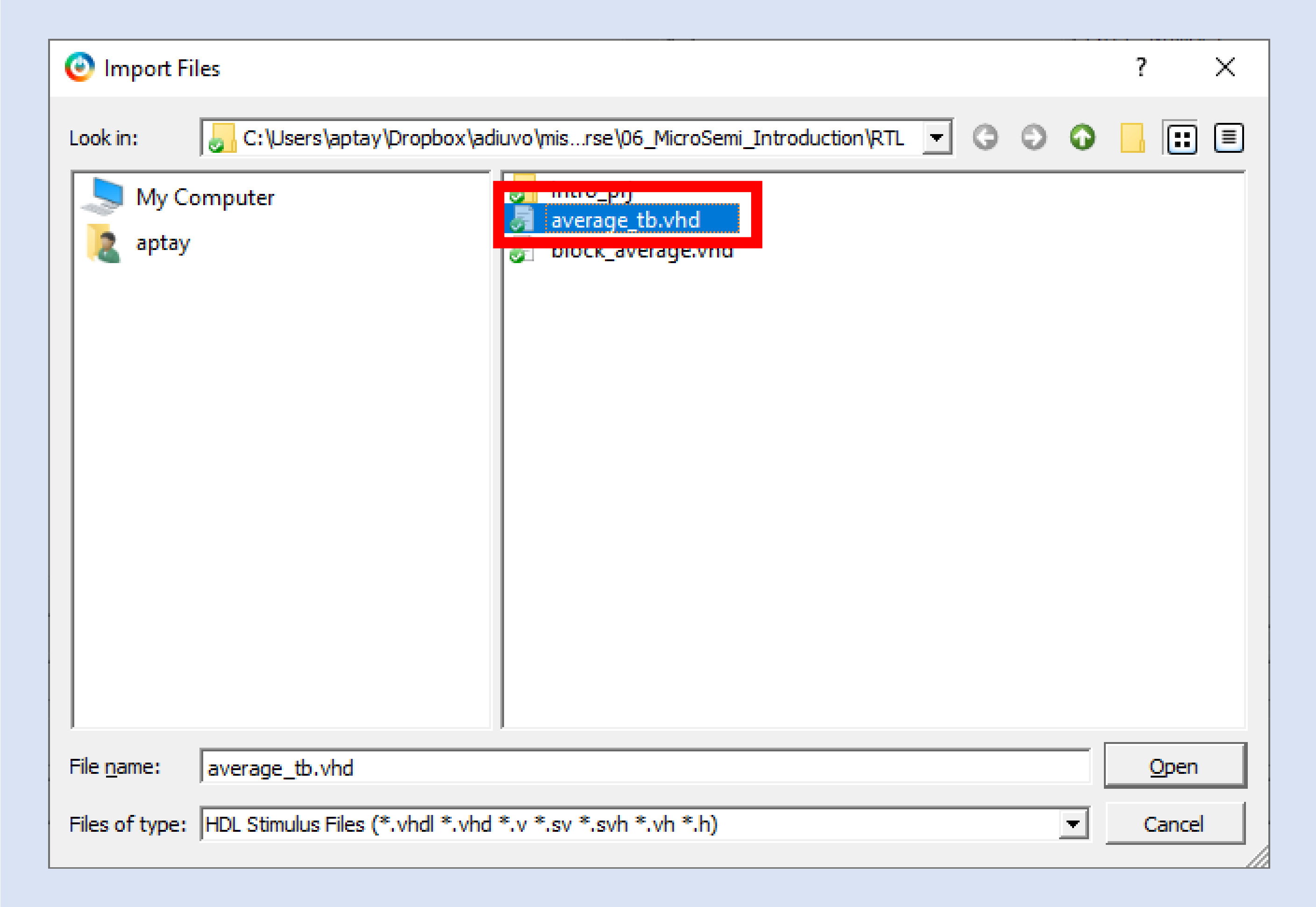
**Step 12 –** Select File, the average_tb.vhd file



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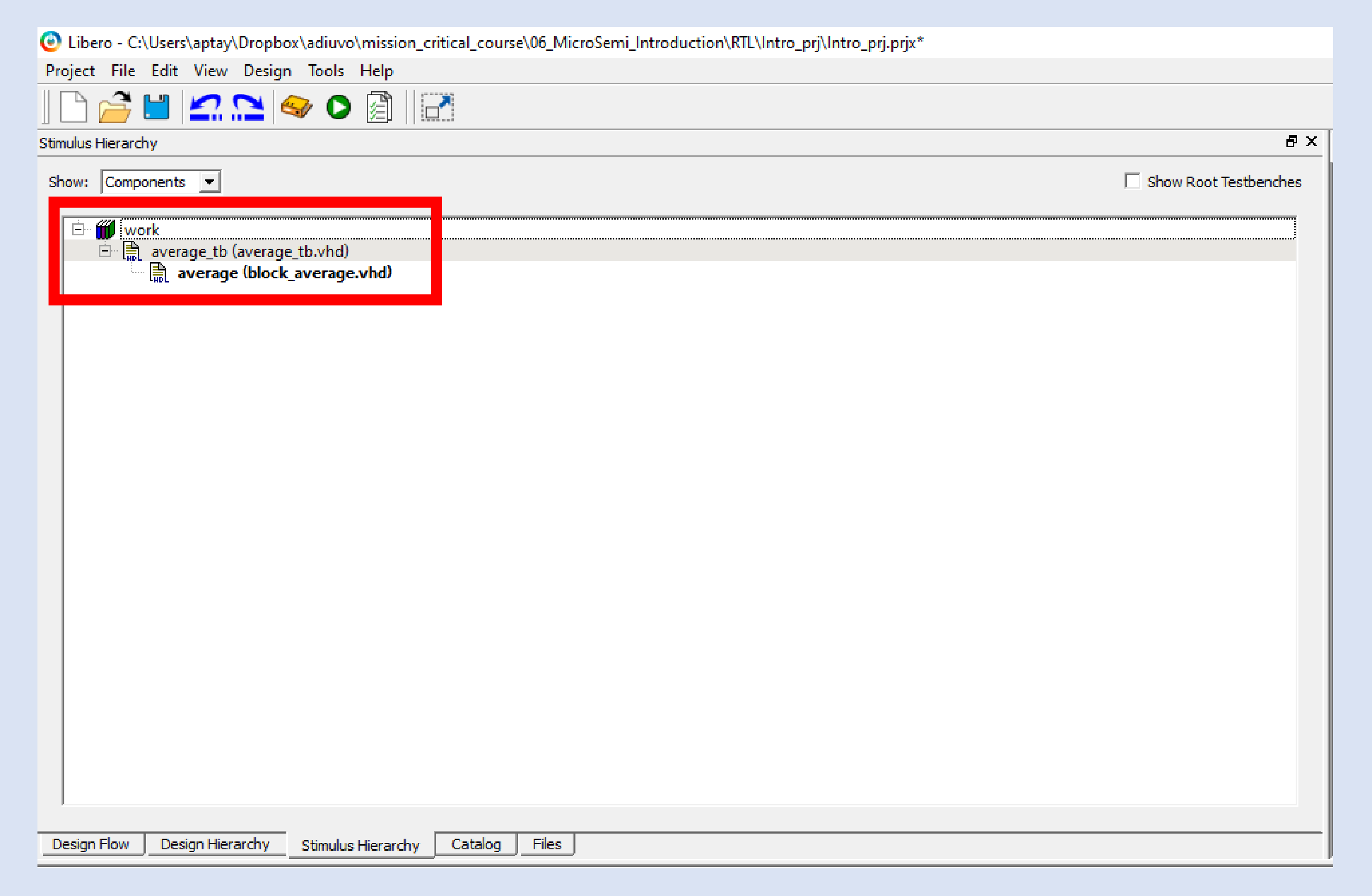
**Step 13 –** The file will show in the Stimulus Hierarchy, expanding the average_tb file should show the block_average RTL file.



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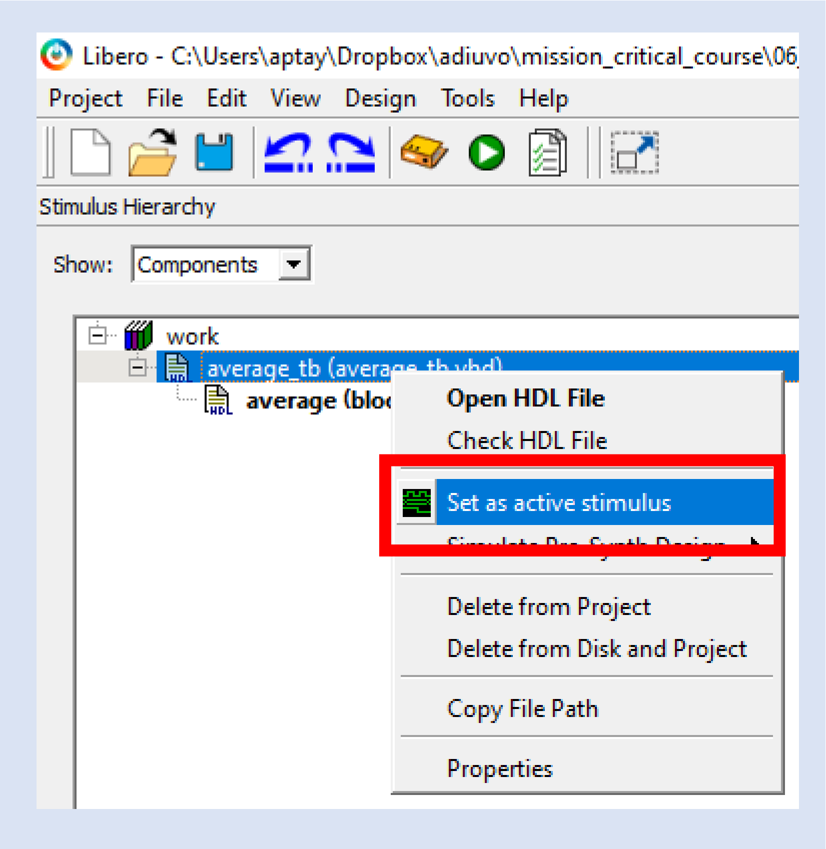
**Step 15 –** Right click on the test bench and select – Set as Active Stimulus



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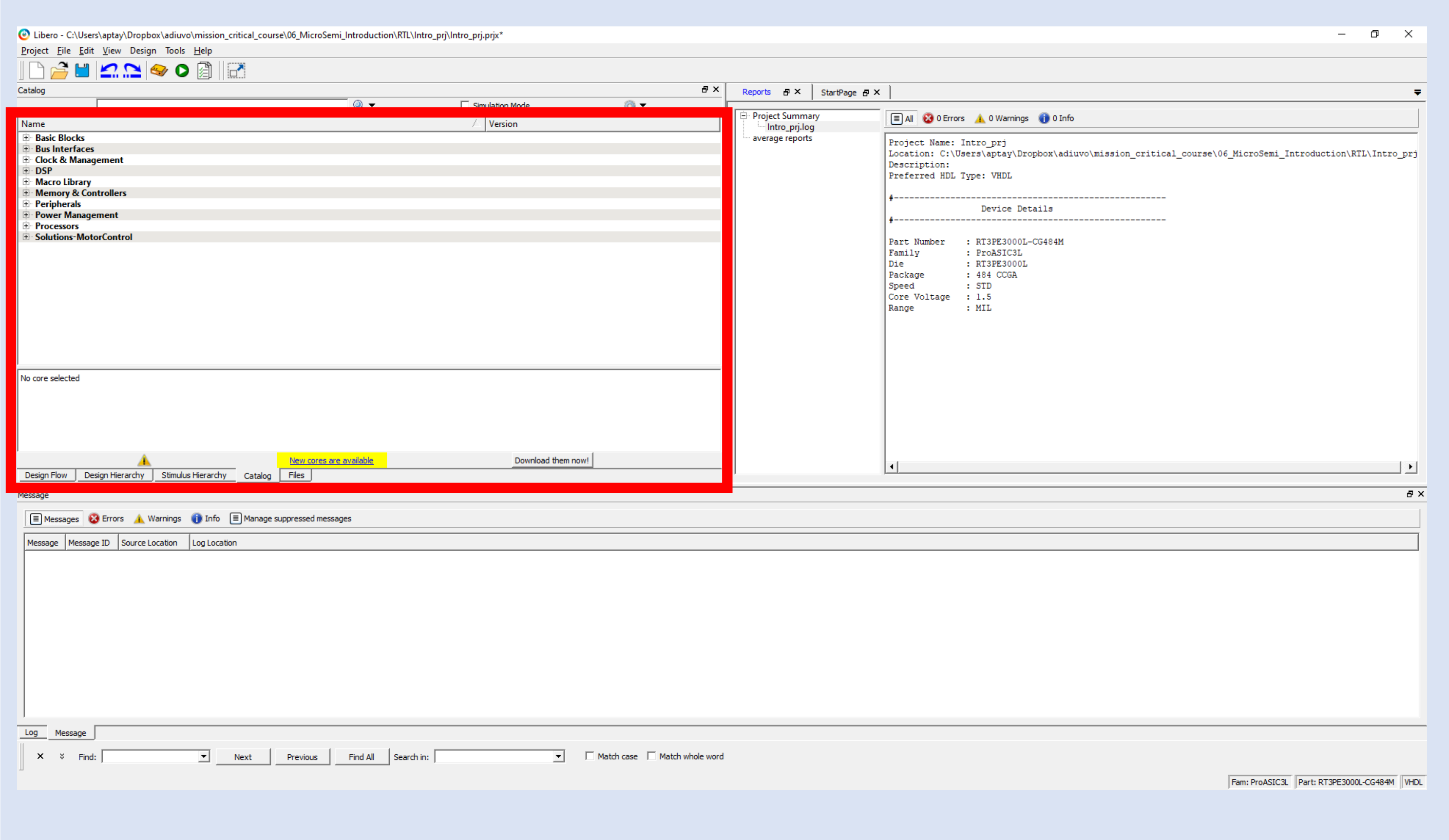
**Step 16 –** Click on Catalog and you will see the IP which can be generated for inclusion in the design



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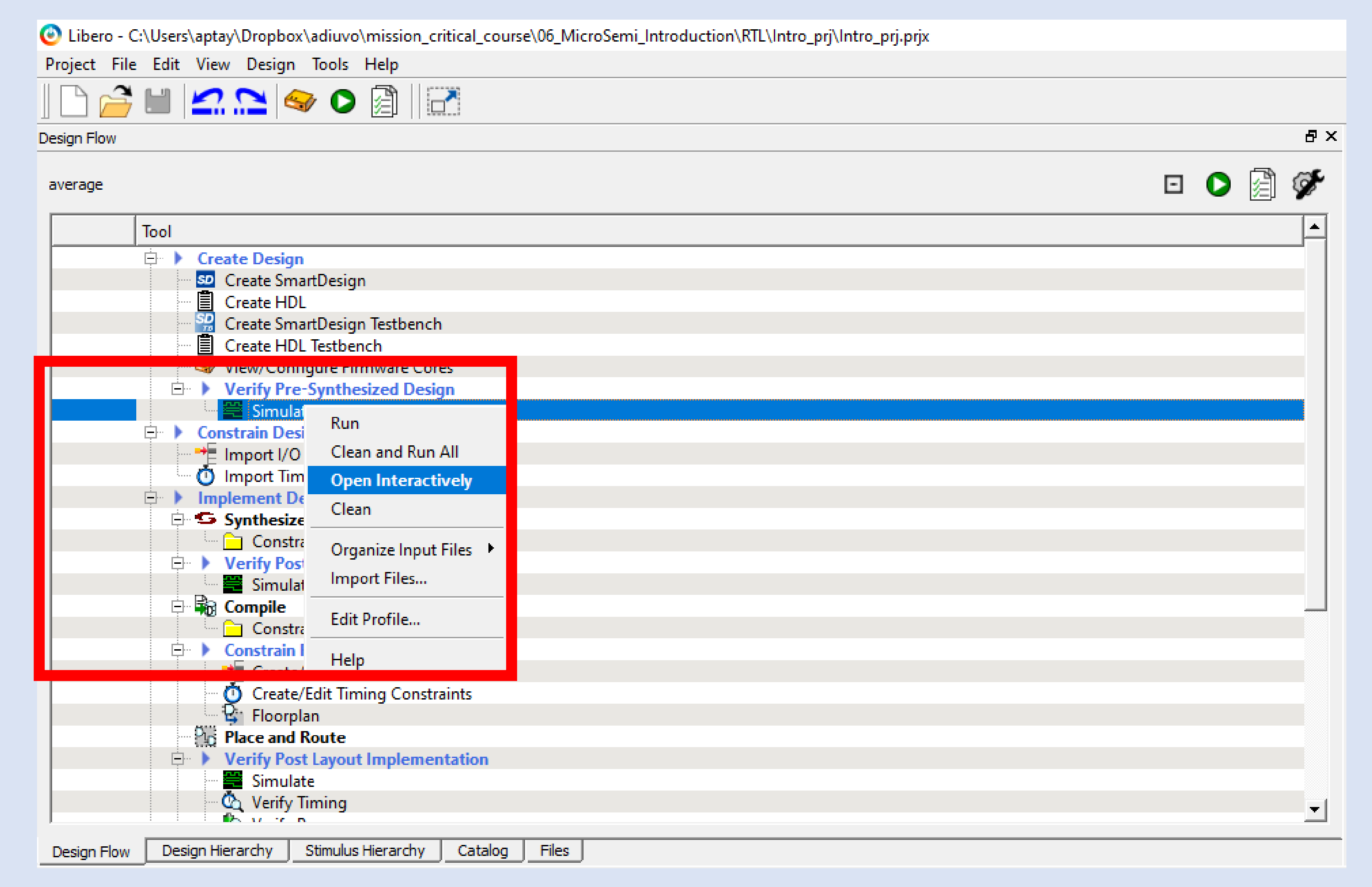
**Step 17 –** Back on the Design Flow Tab, right click on simulate and select Open Interactively



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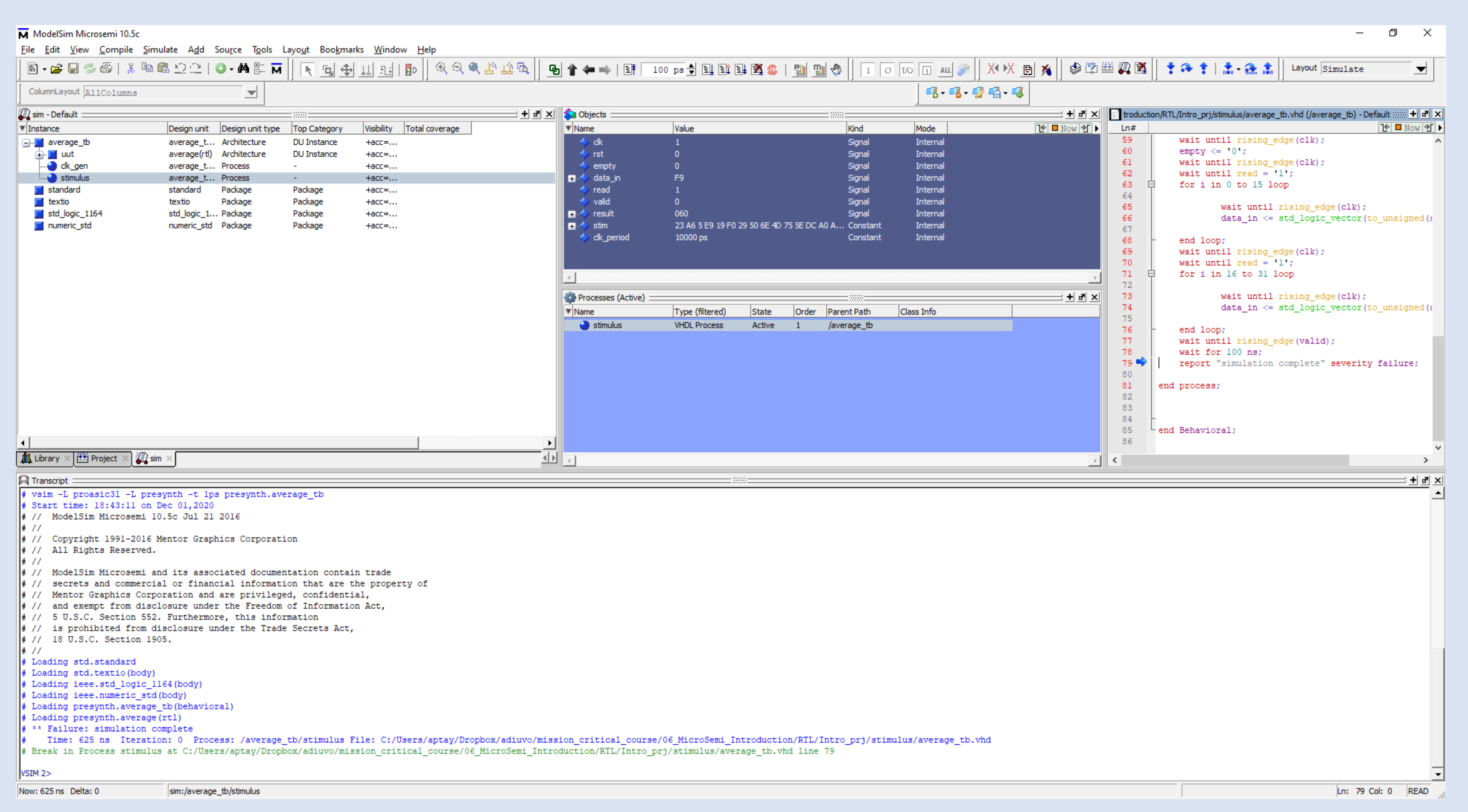
**Step 18 –** This will open ModelSim and run the stimulus file.



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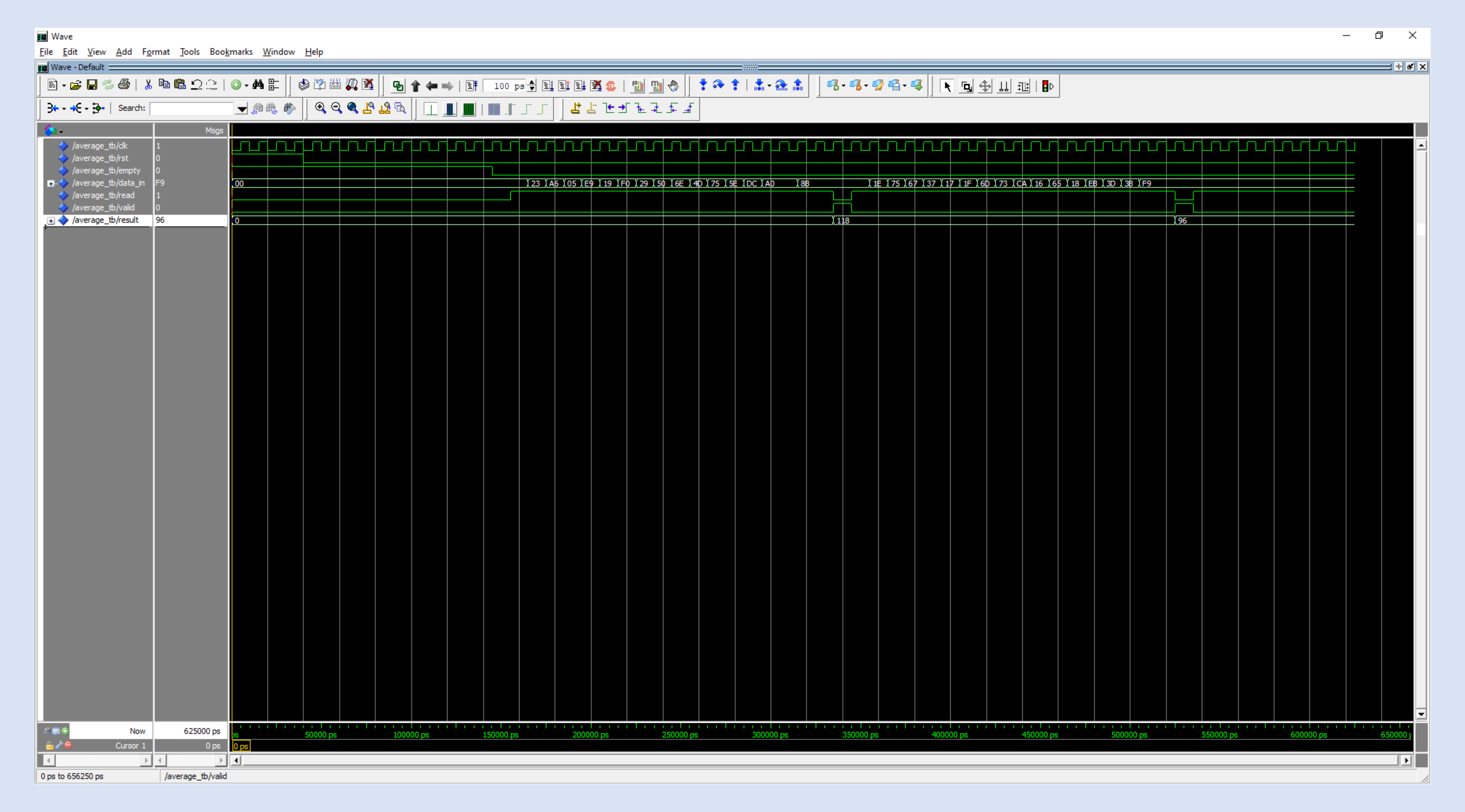
**Step 19 –** Click on the waveform, you will see the average block input and resultant output



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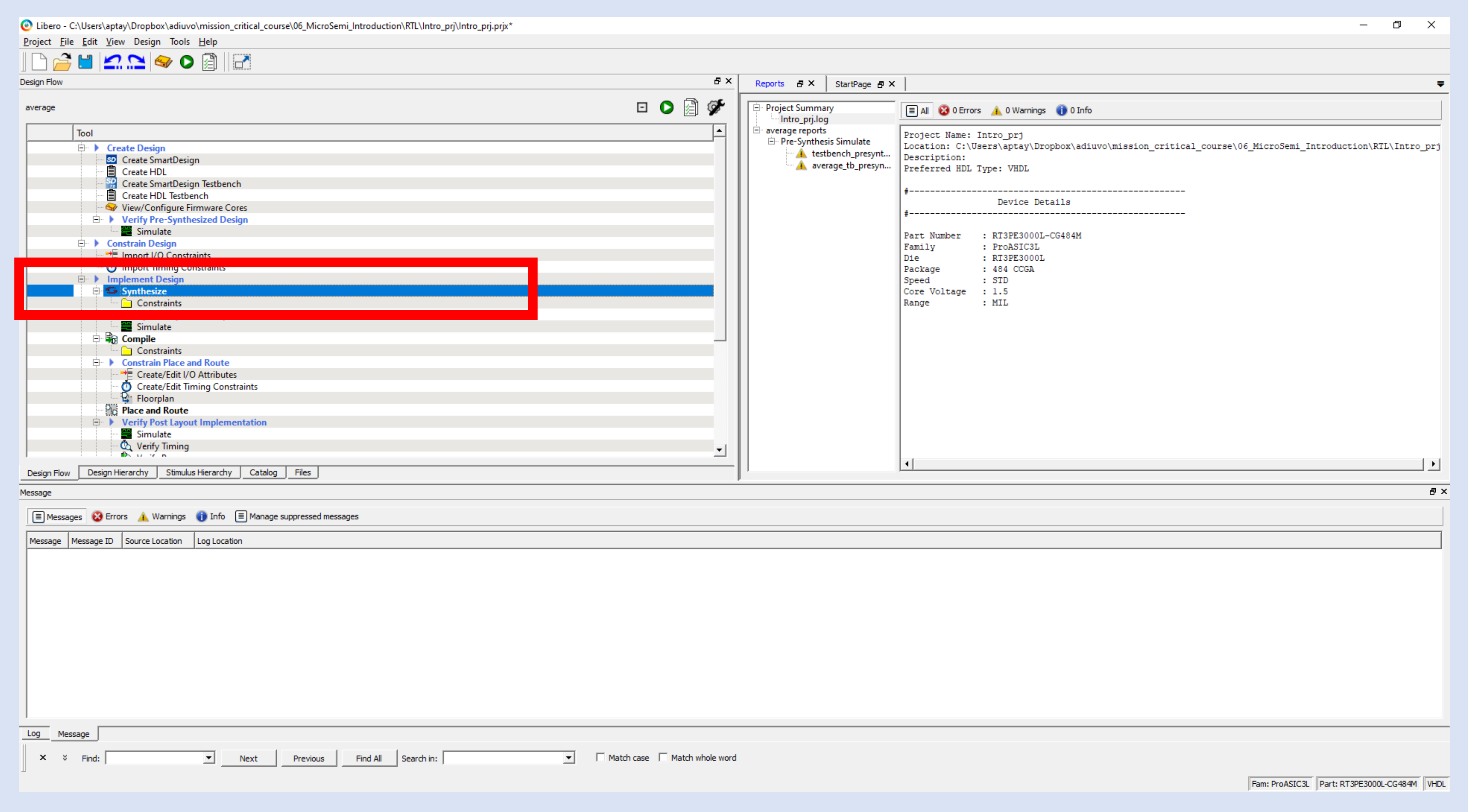
**Step 20 –** Close ModelSim and Return to Libero Project Management Flow – Select Synthesize and Open Interactively



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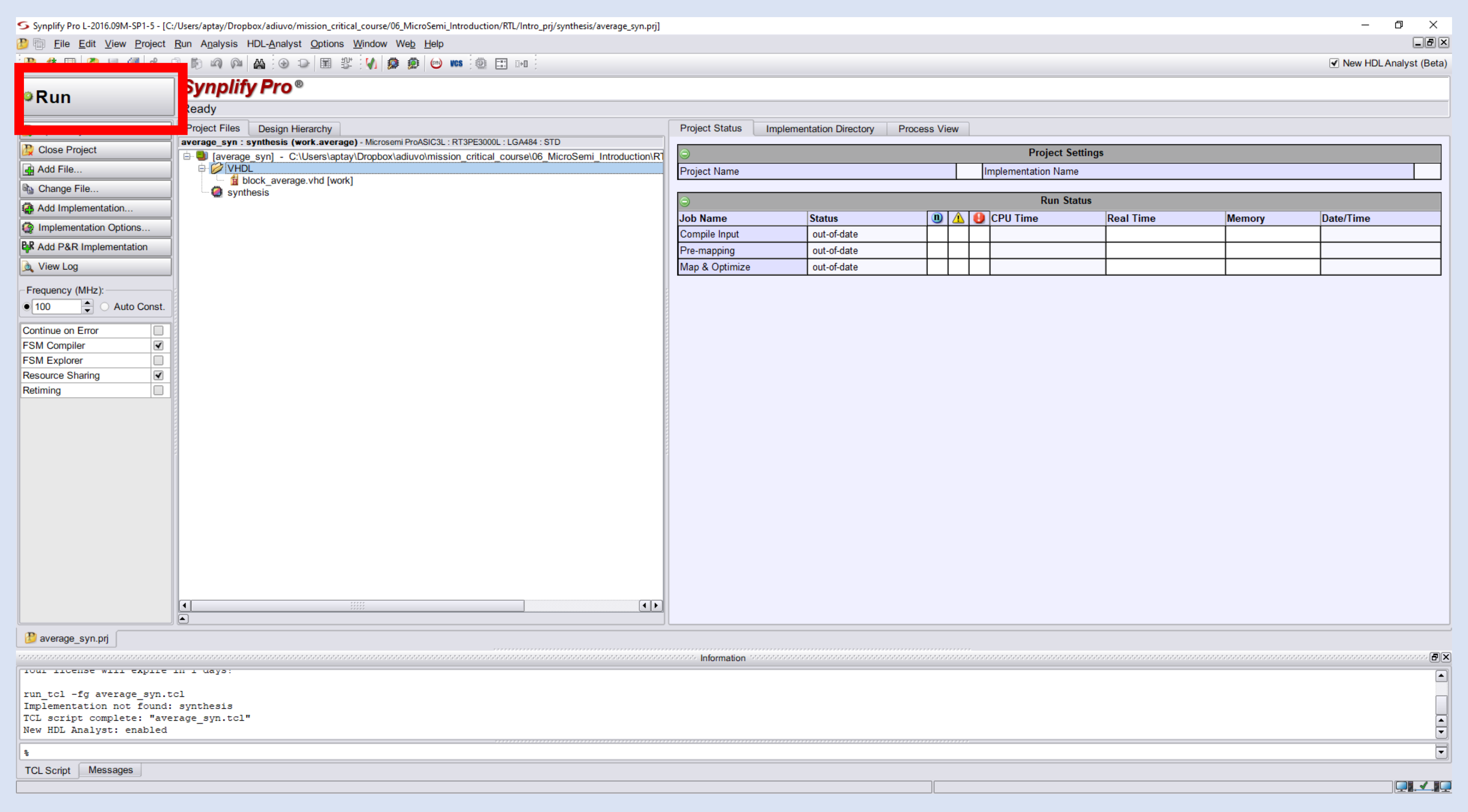
**Step 21 –** Once Synplify Pro opens, expand the VHDL folder and check the block average file is present – Click on run this will synthesis the design



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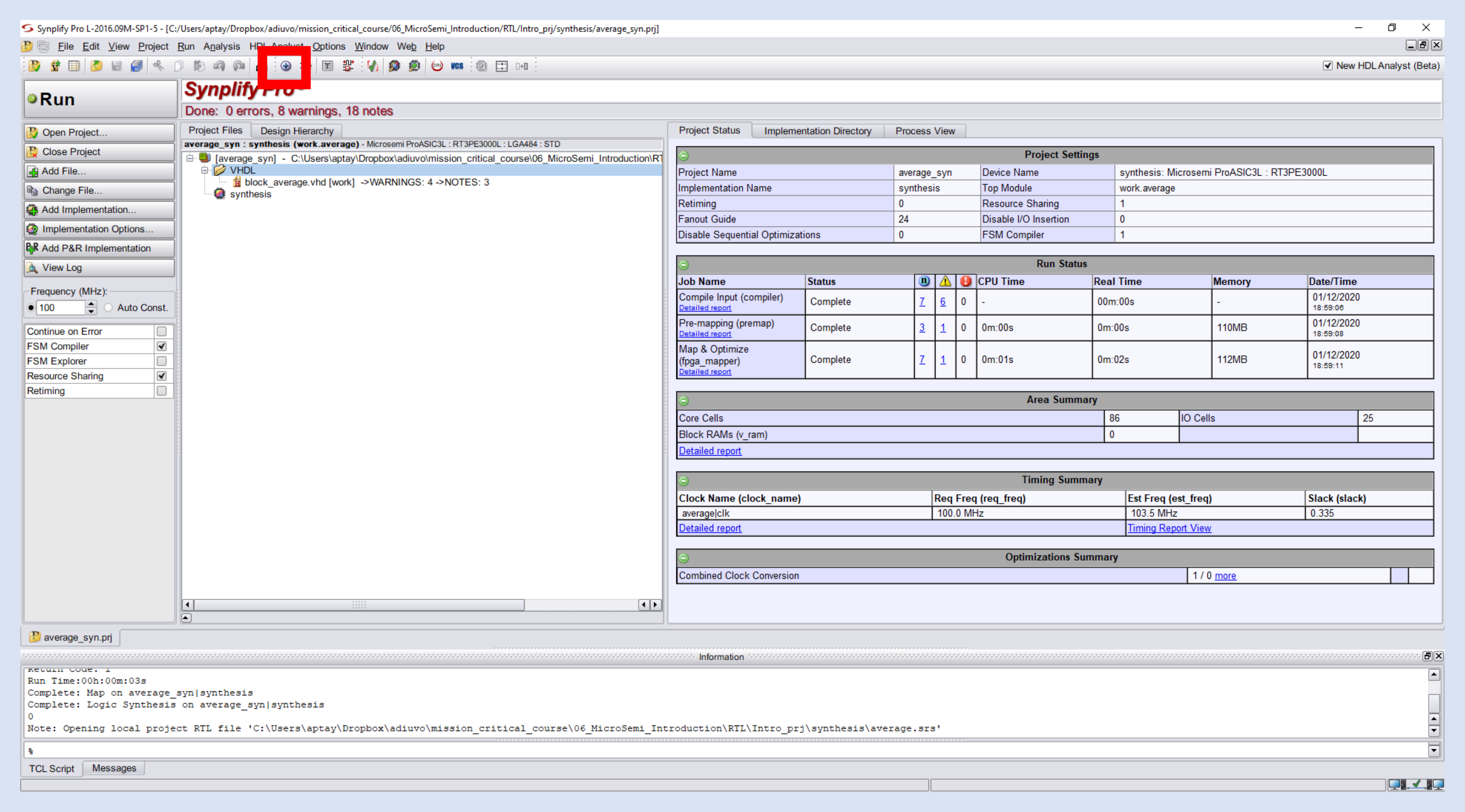
**Step 22 –** Once the Synthesis is completed check there are no errors and open the RTL schematic



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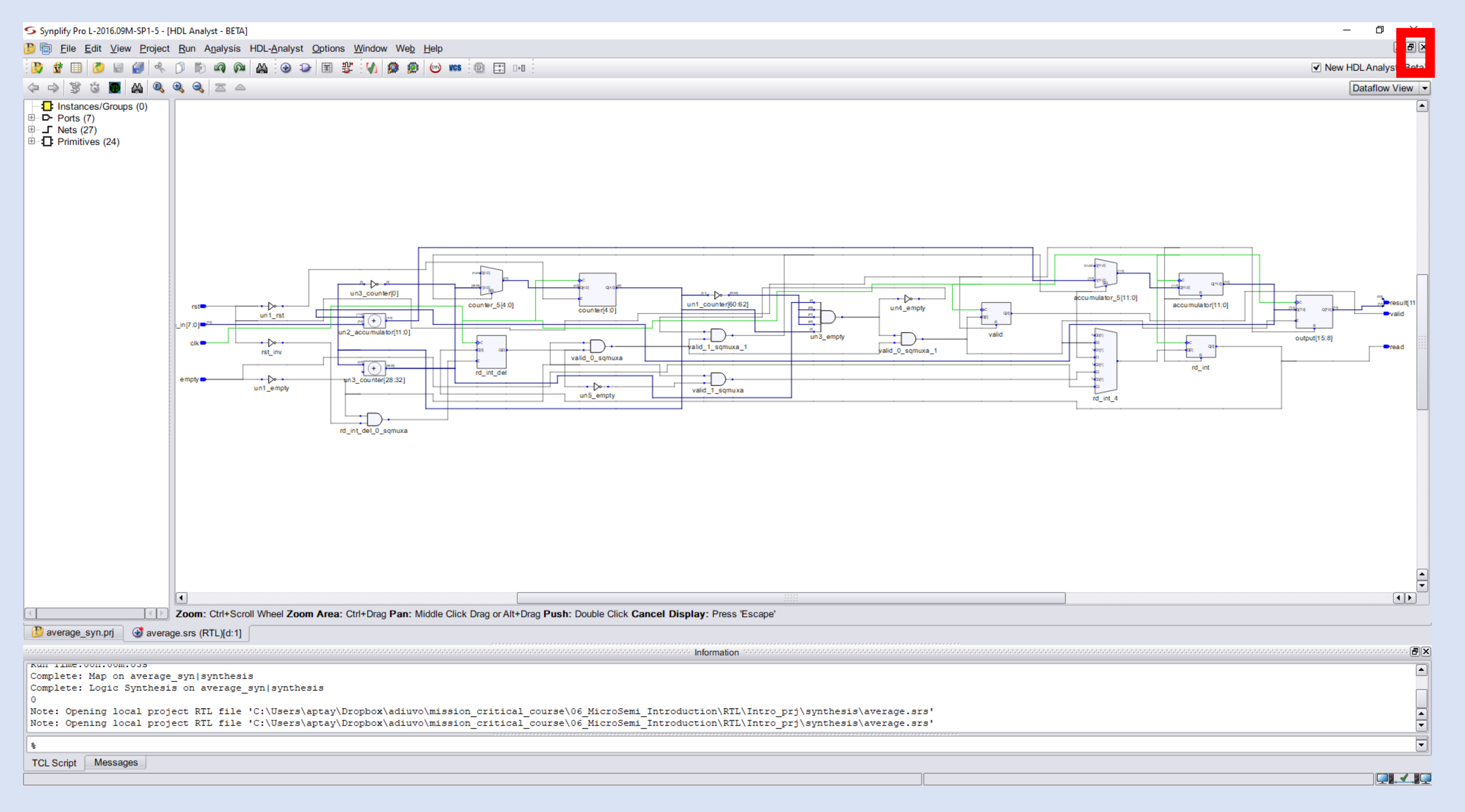
**Step 23 –** The RTL View will show the RTL netlist of your design – This can be useful later when we are looking at implementing high reliability solutions. Close the RTL View.



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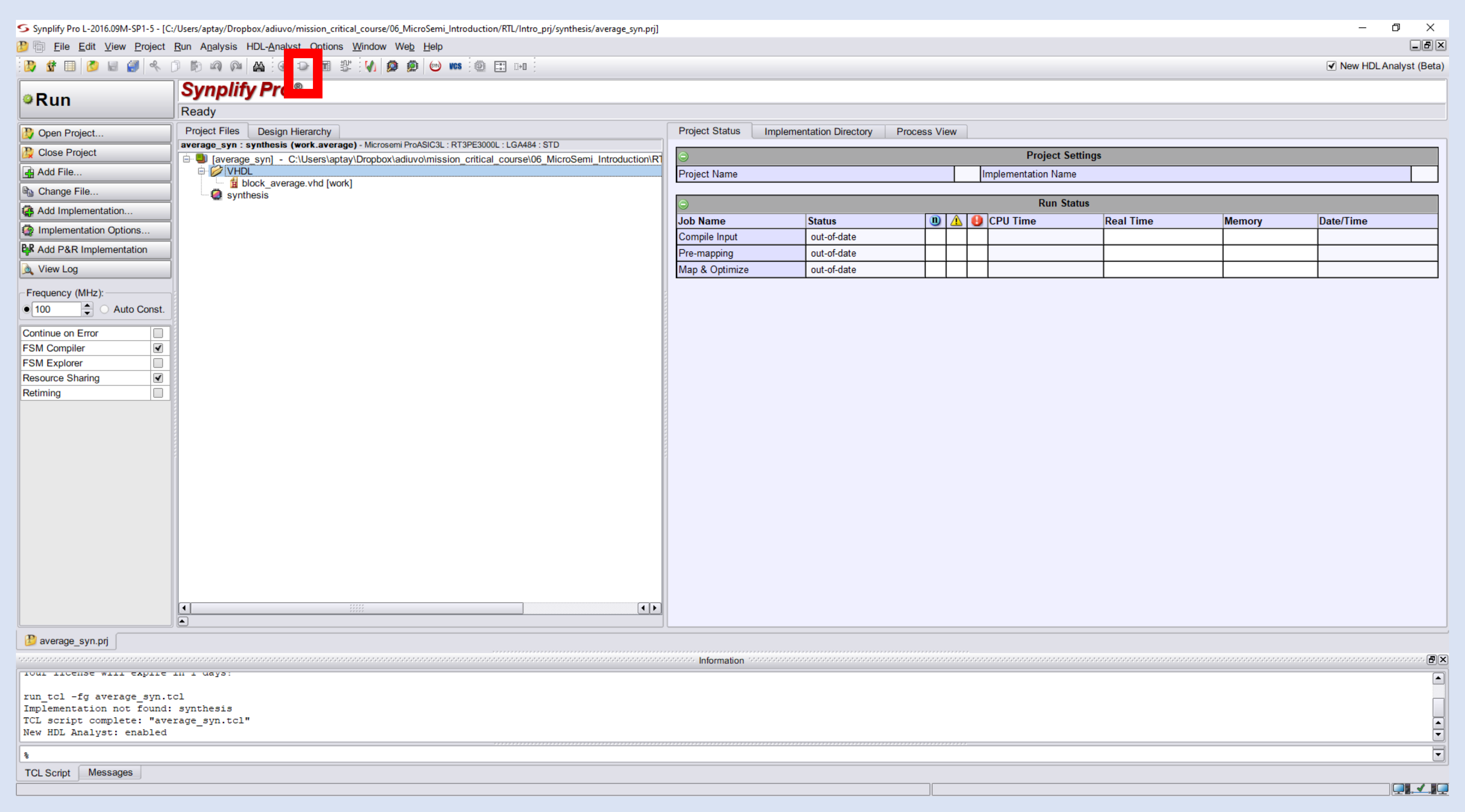
**Step 24 –** Open the Technology View.



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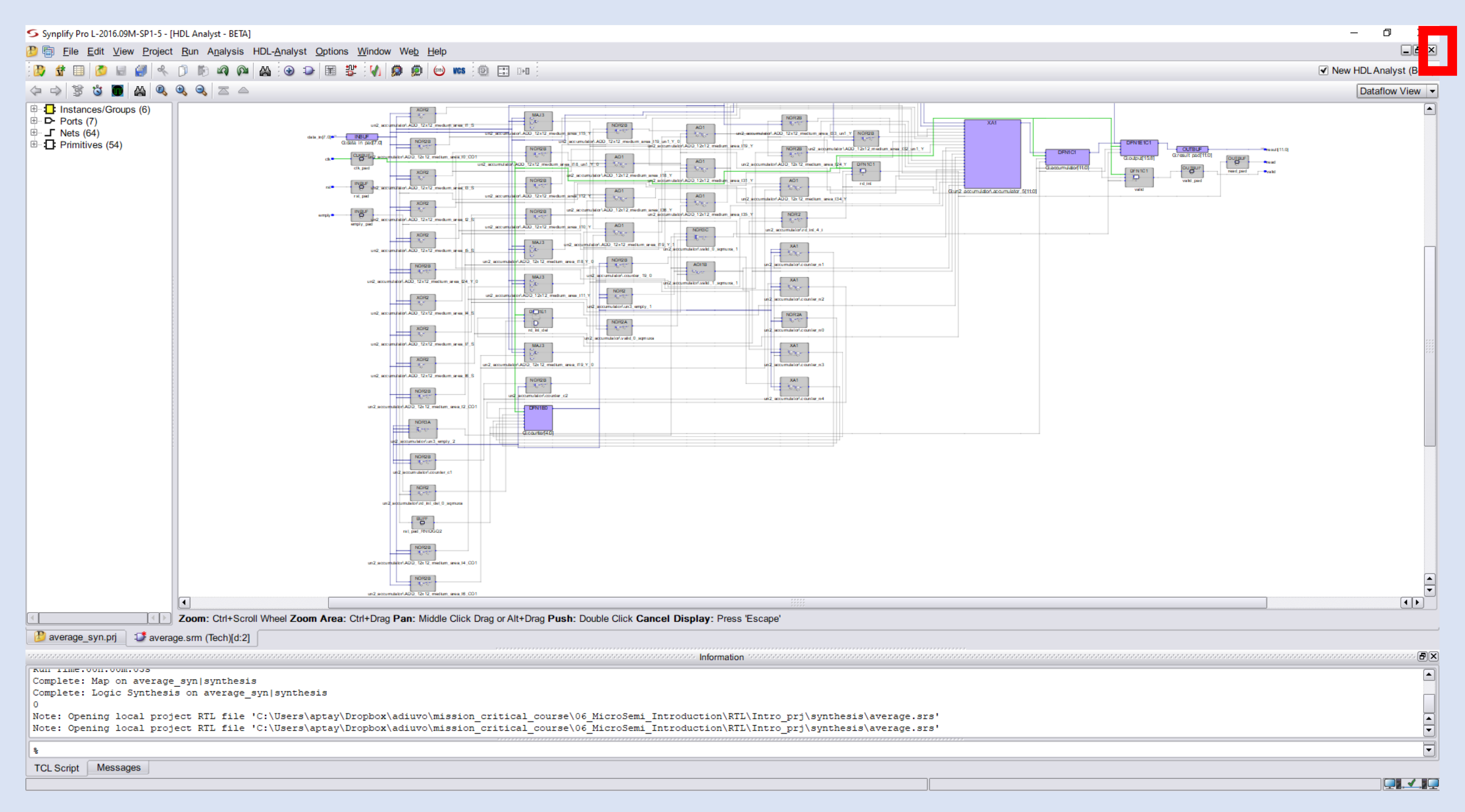
**Step 25 –** The Technology View shows the implemented logic resources in the target device. This view is again useful when investigating issues in synthesis for high reliability designs



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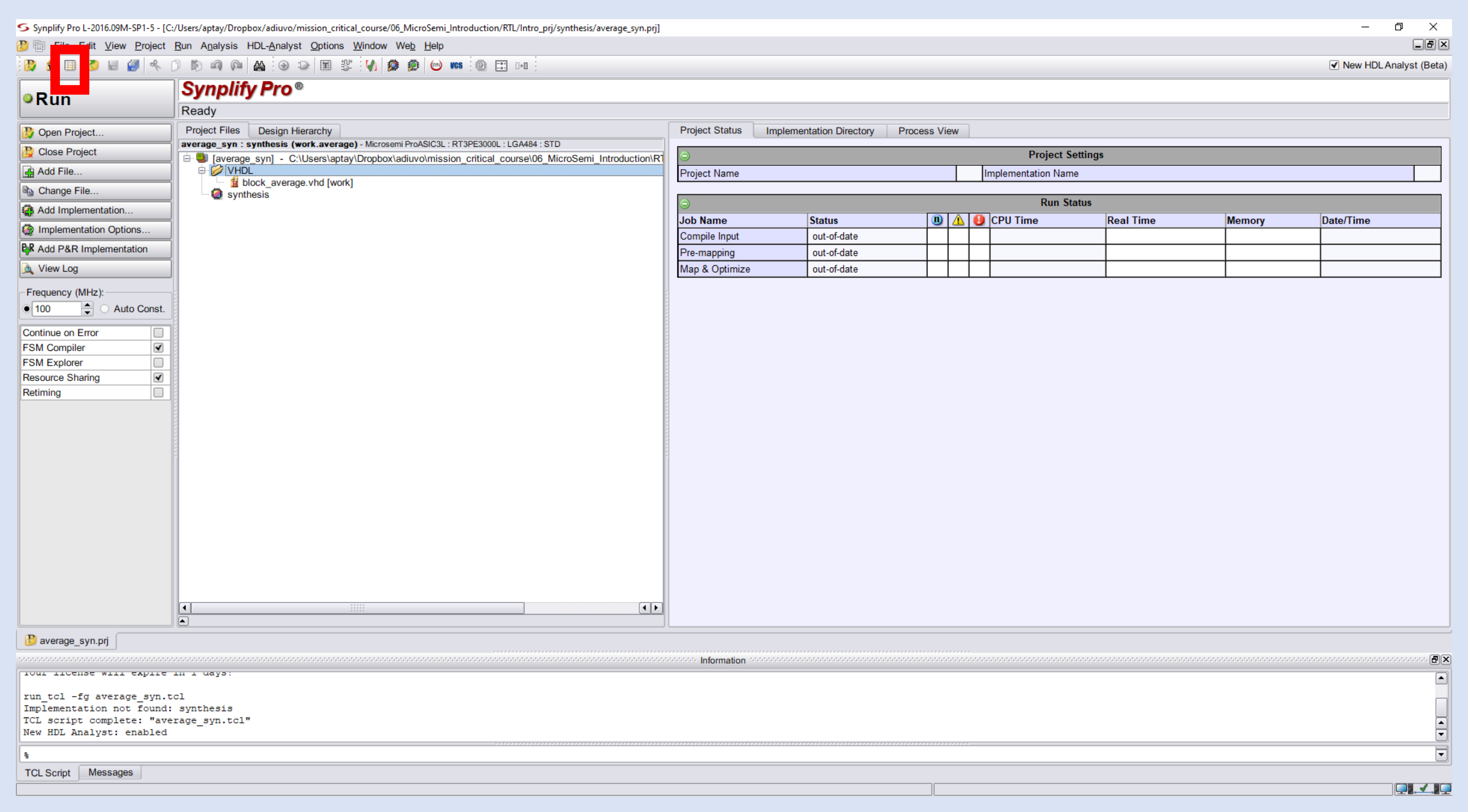
**Step 26 –** If we need to add synthesis constraints e.g. design TMR we can do this via the SCOPE editor



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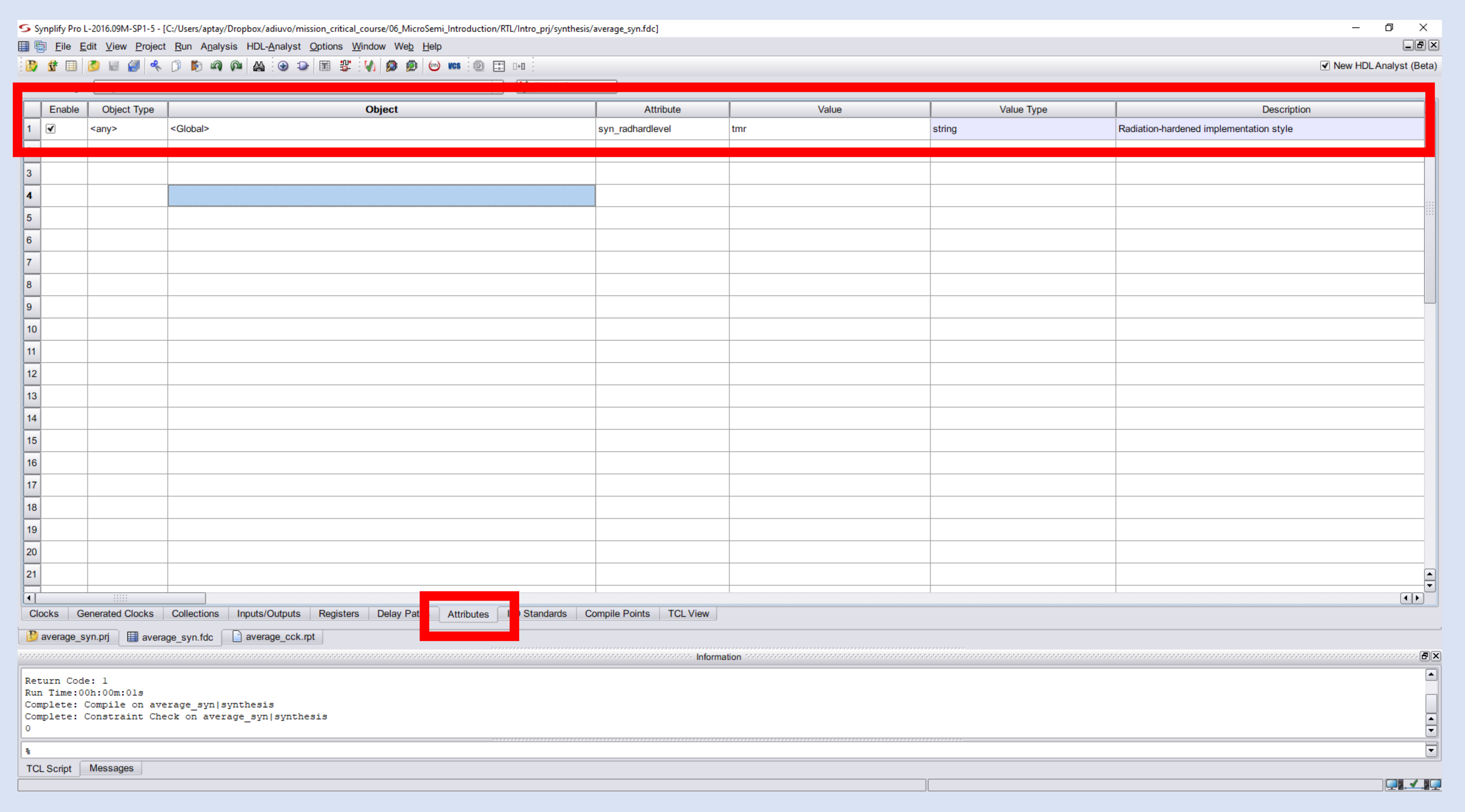
**Step 27 –** Click on the Attributes Tab and select a Syn_RADHARDLEVEL of TMR



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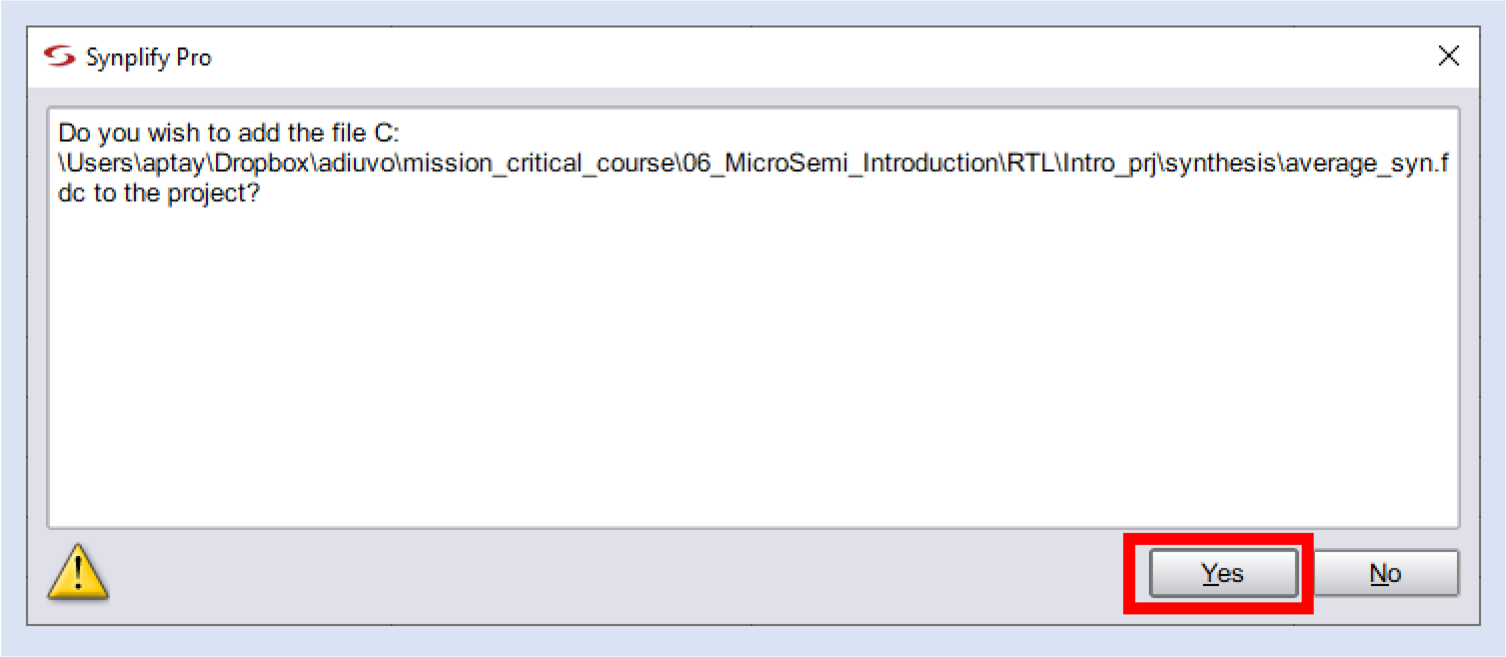
**Step 28 –** Save the file and when prompted add in the saved file to the project



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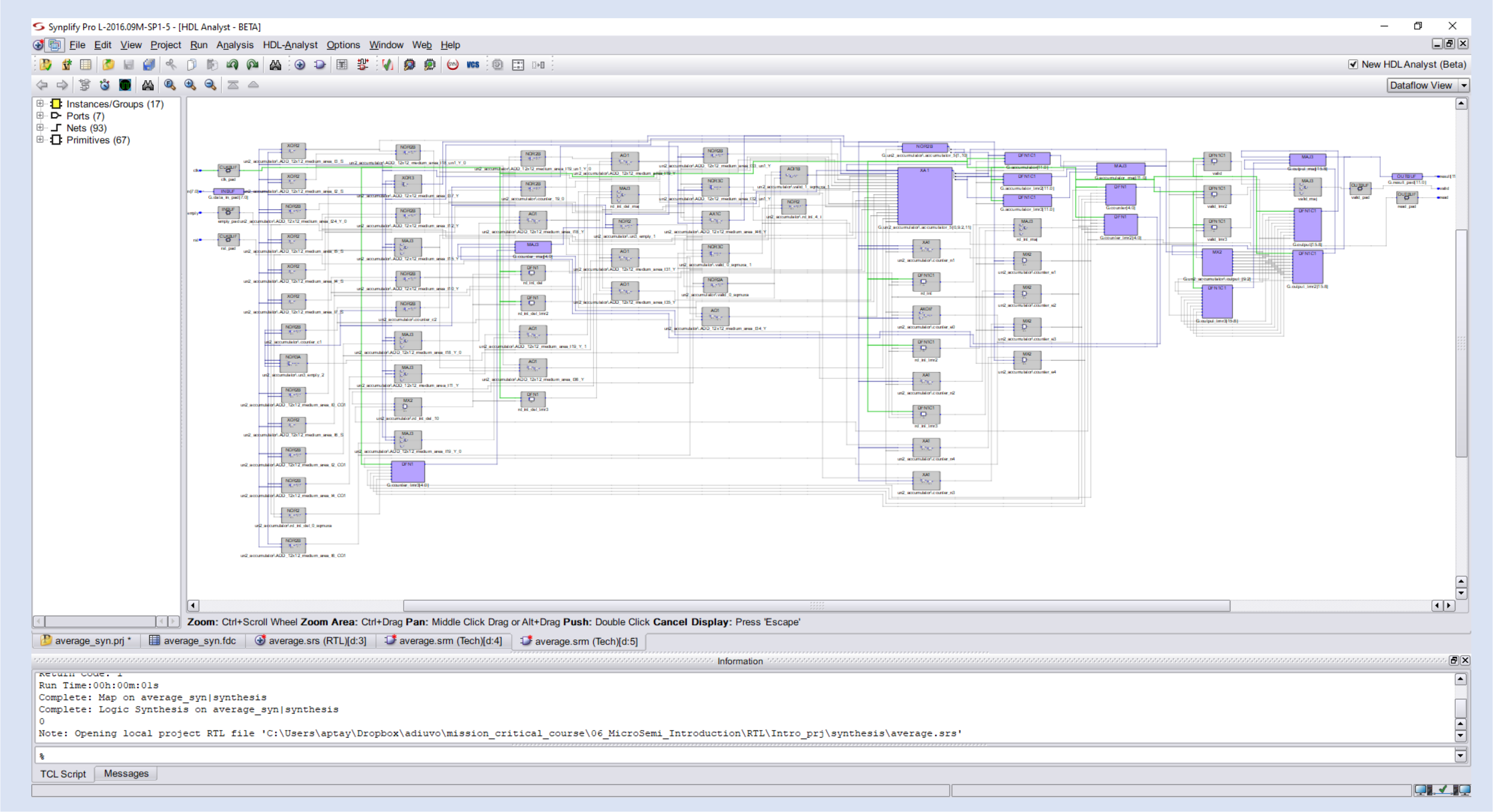
**Step 29 –** Re-Run the Synthesis and examine the Technology view this will present TMR implementation. Explore the design then close Synplify and return to Libero



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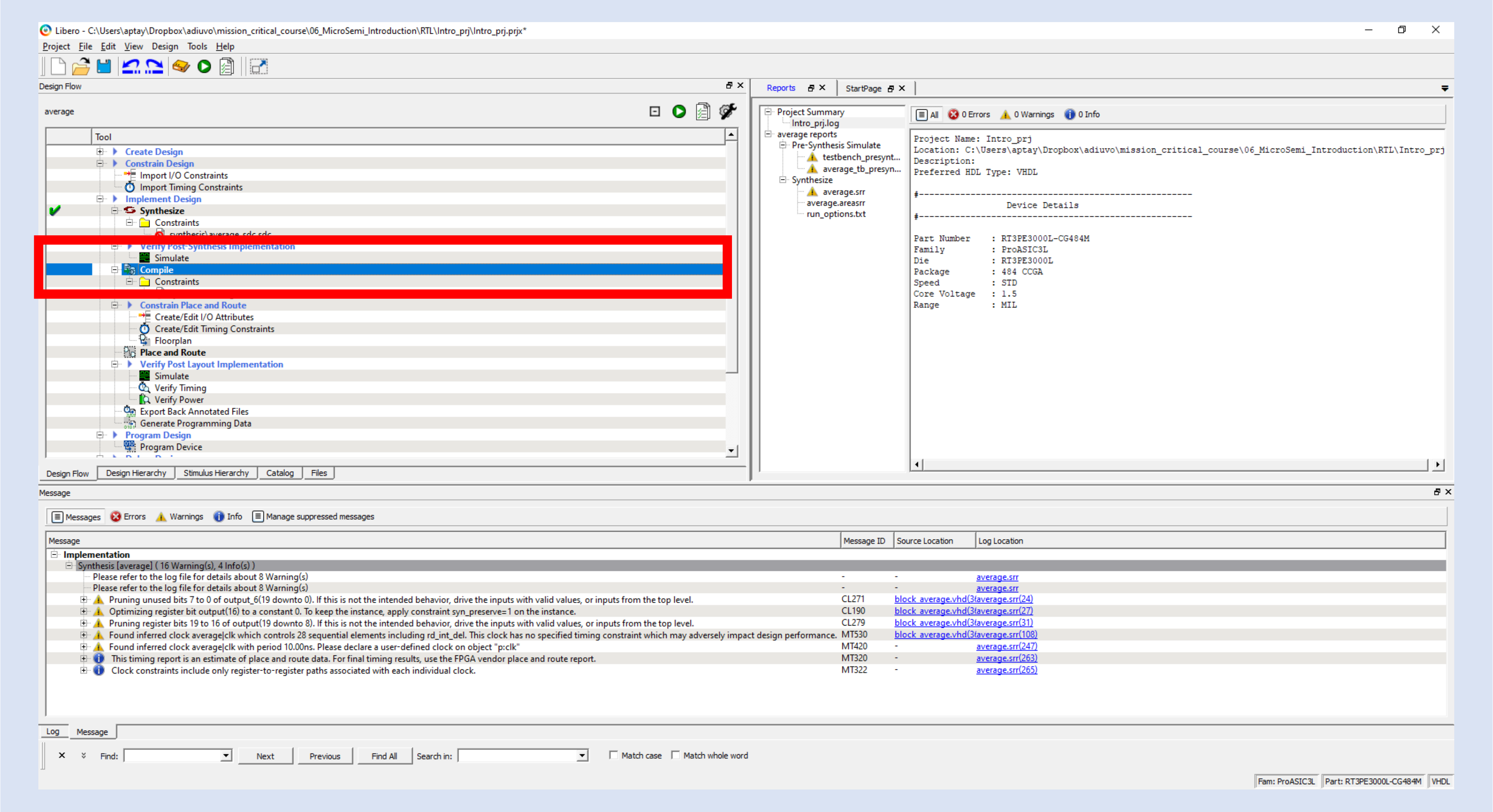
**Step 30 –** Back in Libero select the Compile and again right click and open interactively



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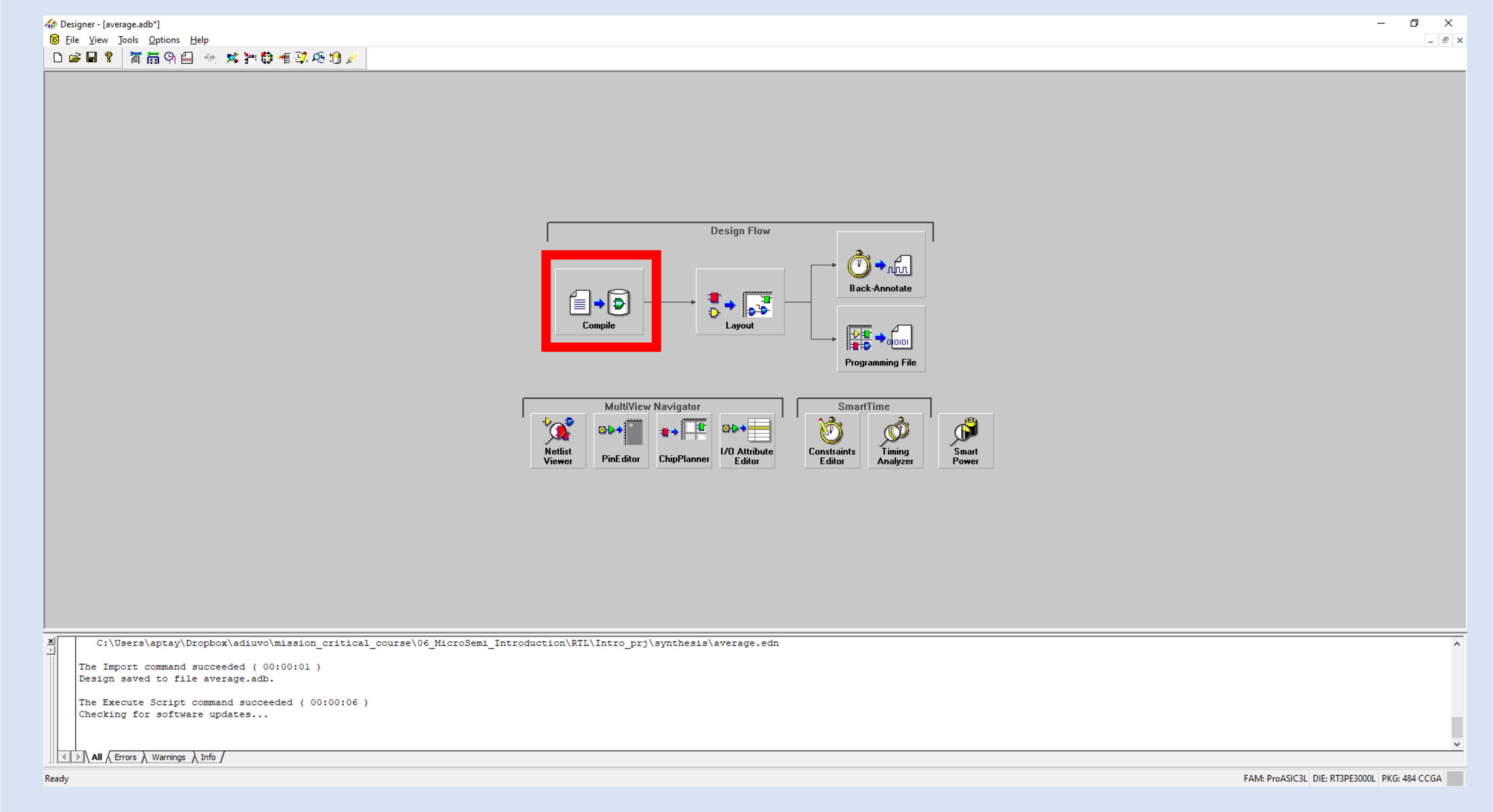
**Step 31 –** In Designer click on Compile - If a compile option dialog appears leave unchanged and click OK



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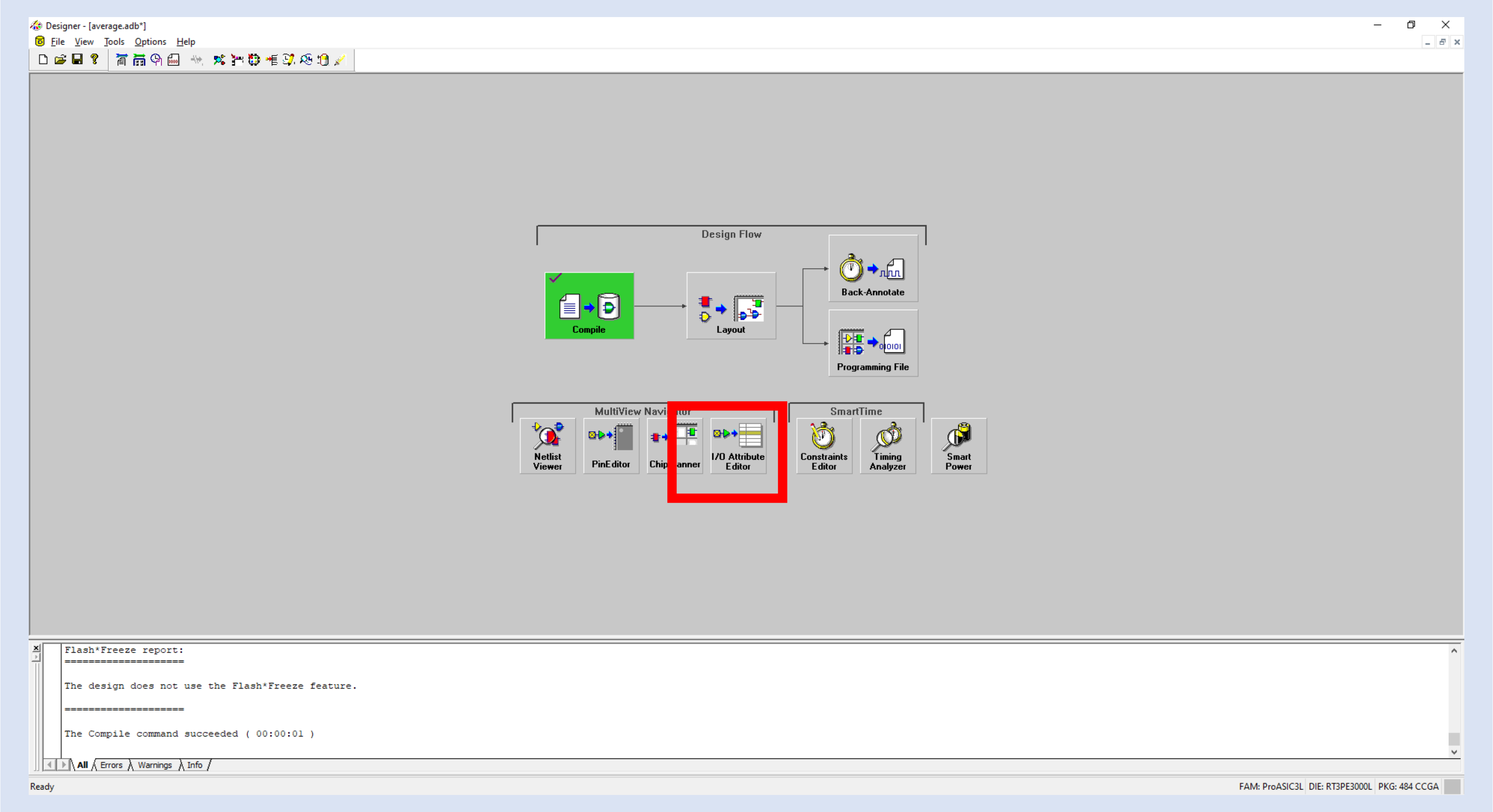
**Step 32 –** Click on I/O Attribute Editor



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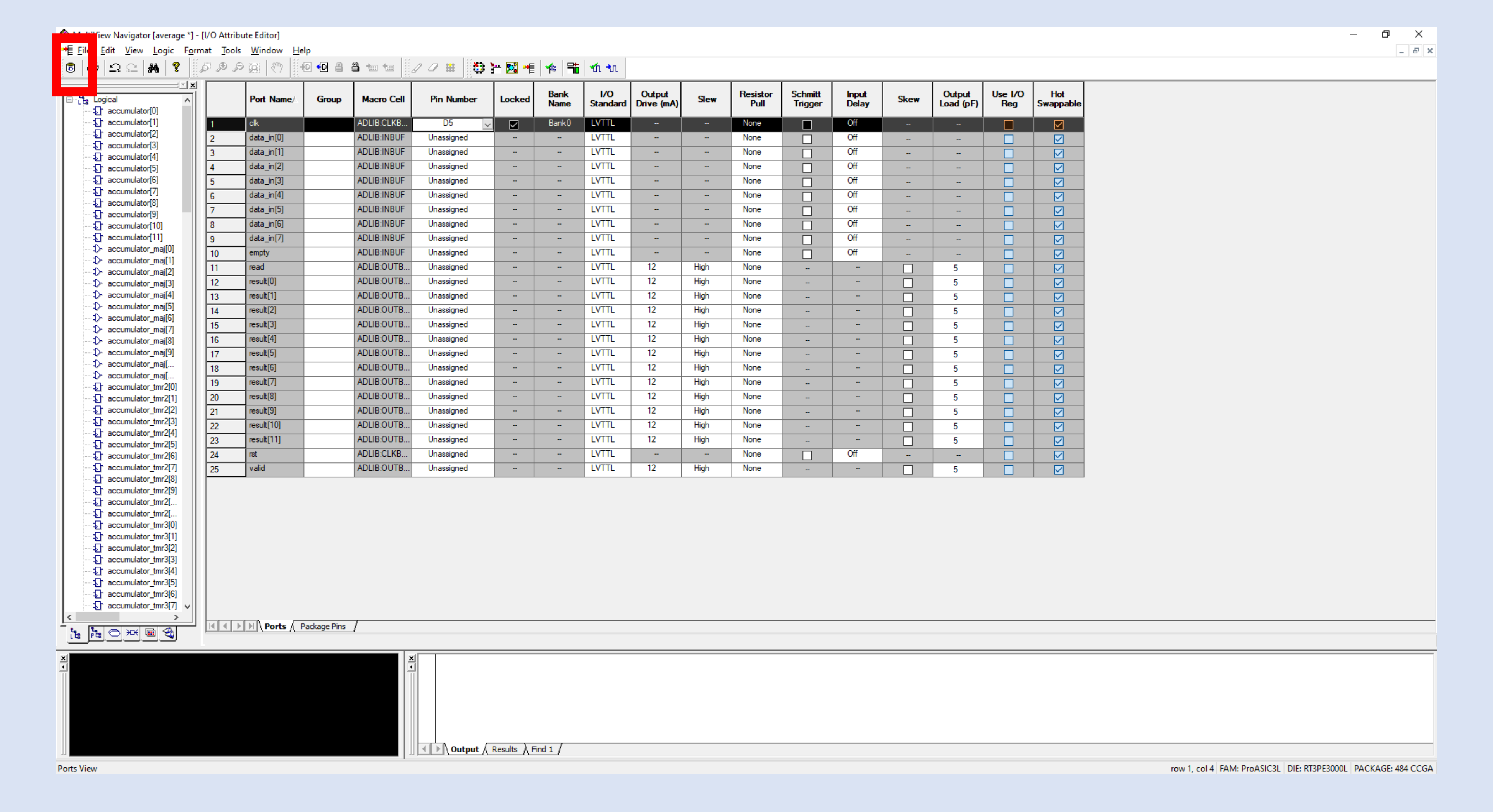
**Step 32 –** Click on I/O Attribute Editor – Make sure clock is on D5



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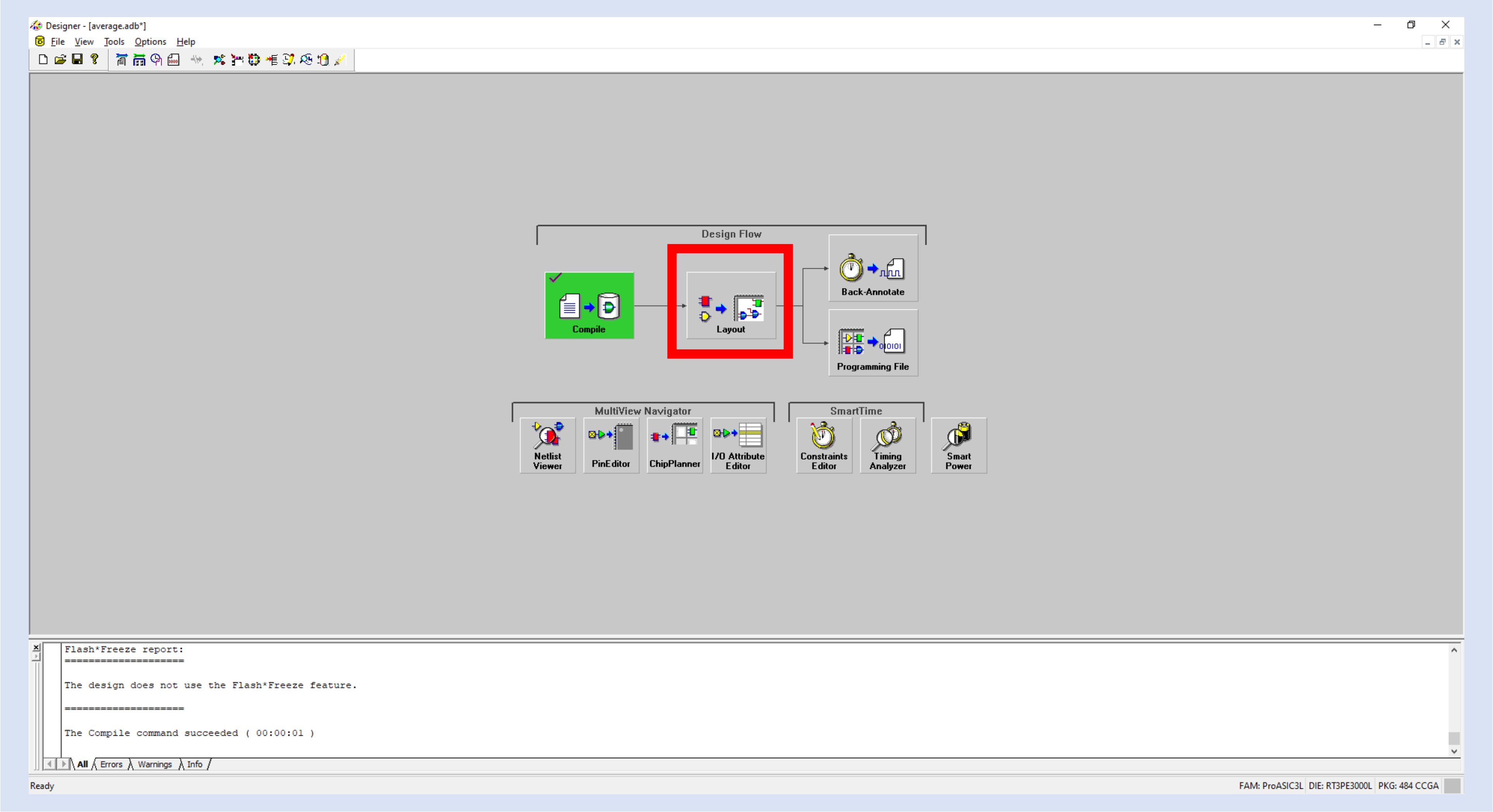

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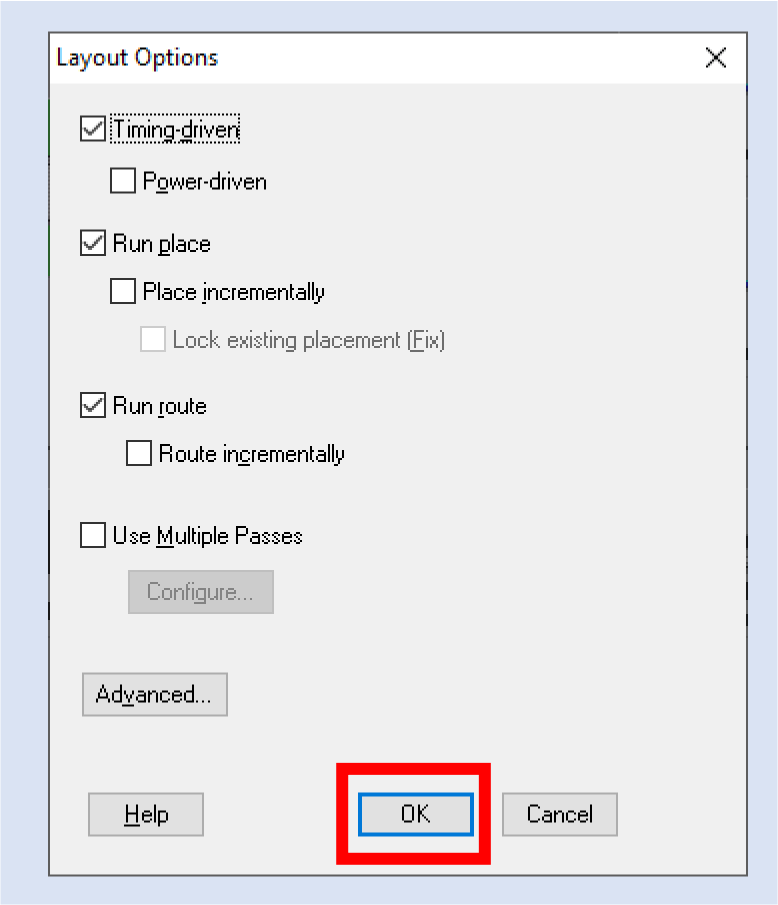

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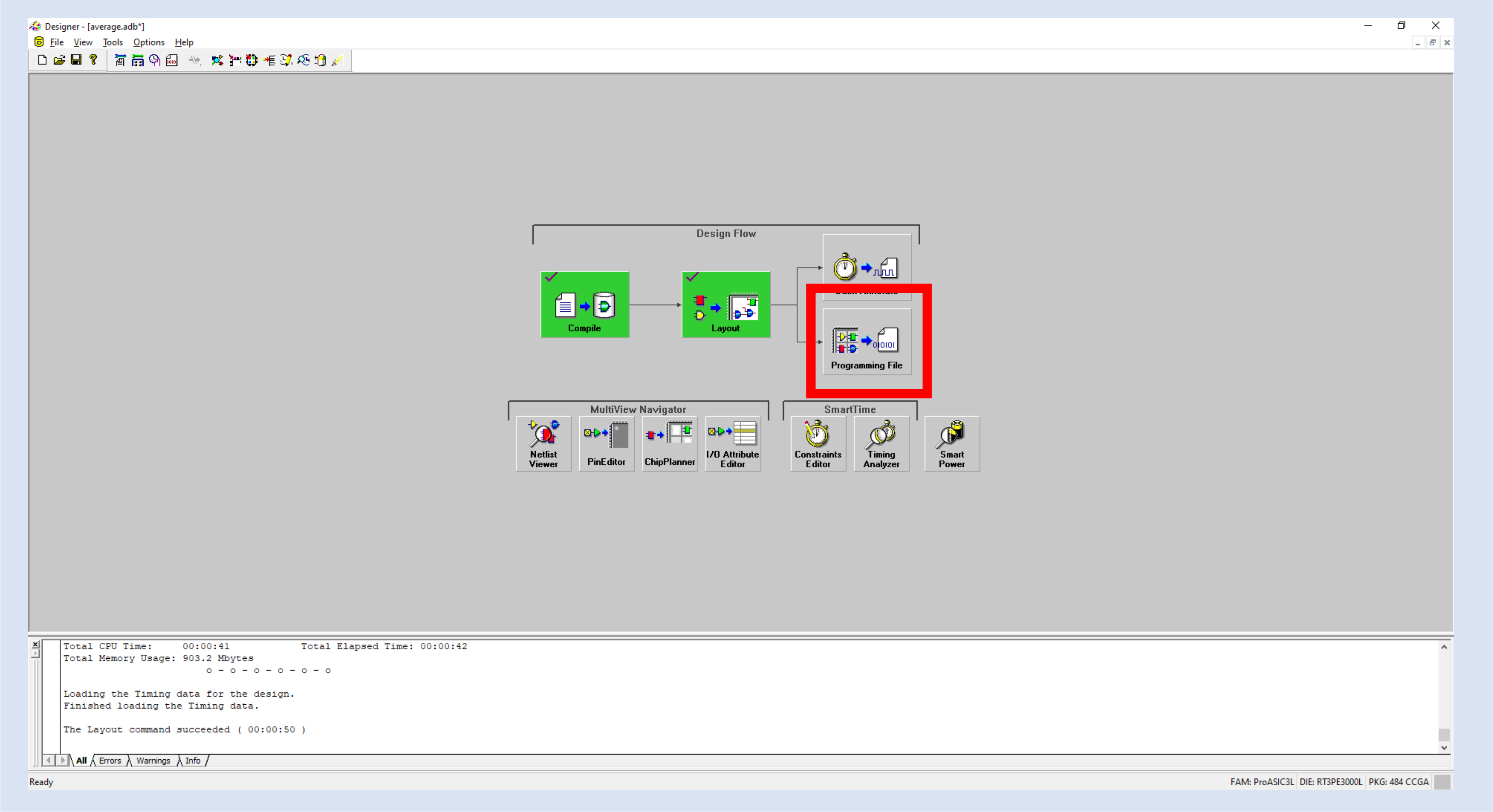

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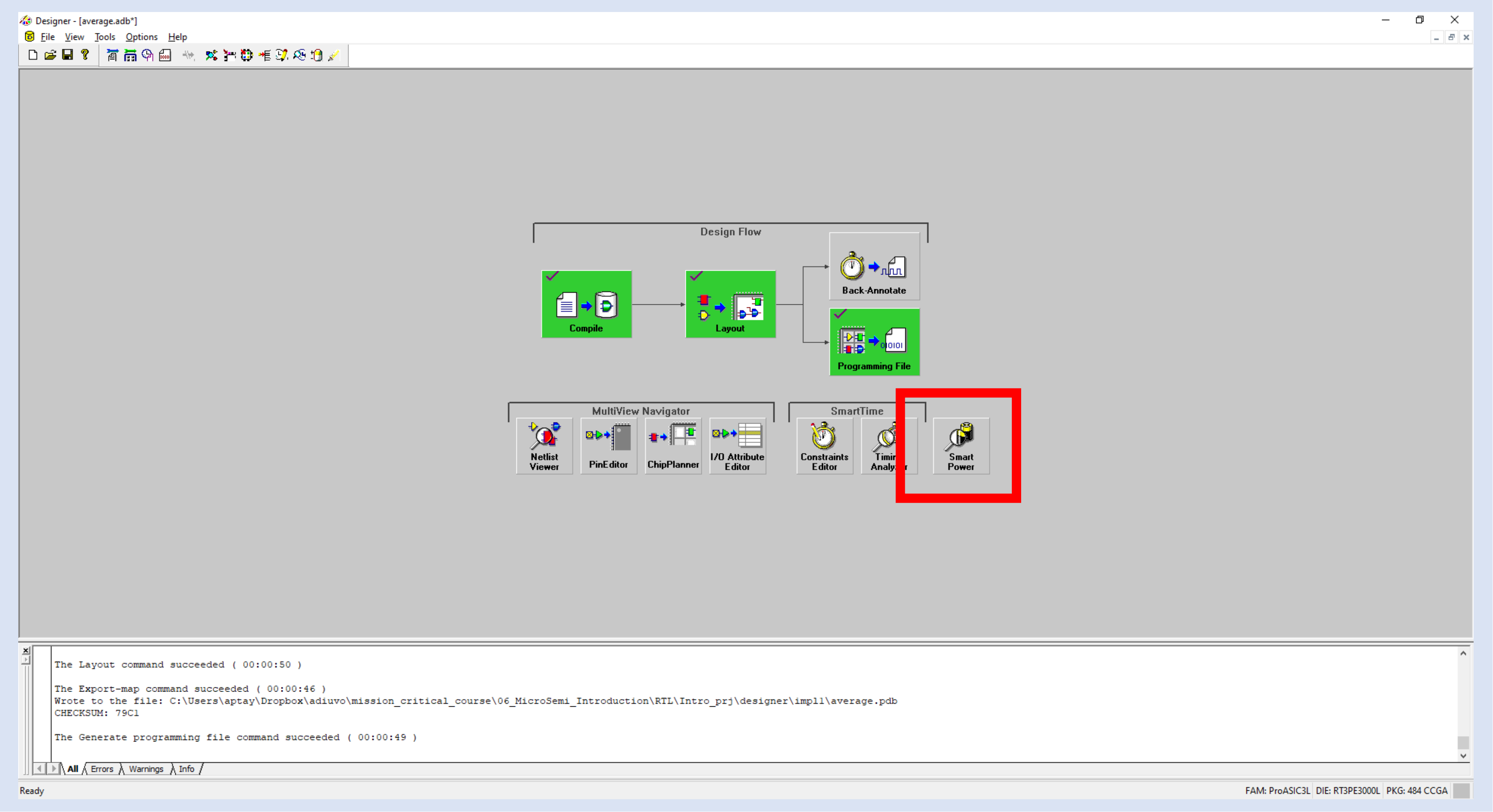

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